

公平与效率的融合：来自GRI与SASB标准的证据

摘要

我们对于采用GRI或SASB作为ESG信息披露标准的理论影响提出了独到的见解，从而对关于ESG双重或单一重要性的问题，即“公平与效率”的博弈问题做出了贡献。我们的主要贡献是，通过对ESG报告的文本分析，建立了第一手的、公司层面的ESG信息披露标准数据库。此外，我们还建立并实证检验了ESG反馈机制的四大渠道，即披露成本、ESG评级差异、间接融资和碳排放如何受到“公平和效率”博弈，或者说不同信息披露准则的显著影响。最后，我们给出了关于标准趋同的建议，即需要将GRI作为主体，适当融入SASB的精神内核，也即在“公平”的同时适当注重“效率”。

关键词： ESG信息披露标准，GRI，SASB，ESG反馈体系，趋同

The Convergence of Equity and Efficiency: Evidence from the GRI and SASB Standards

Abstract

We contribute to the emerging debate on ESG double or single materiality issue, representing the game of equity and efficiency, with original insights about the theoretical implications related to the adoption of the Global Reporting Initiative (GRI) and the Sustainability Accounting Standards Board (SASB) as reporting standards. Our major contribution is to build a first-hand company-level ESG standard database through ESG report textual analysis. Furthermore, we empirically verify how equity and efficiency affect the cost of disclosure, ESG rating divergence, indirect finance and carbon emissions, the four channels of the ESG feedback mechanism which are significantly influenced by disclosure standards. We finally give suggestions on standards convergence, that is, to improve the “efficiency” of the GRI Standards, which represents “equity”.

Keywords: ESG disclosure standards, GRI, SASB, ESG feedback mechanism, Convergence

JEL classifications: M14, G24

1 Introduction

Environmental, social, and governance (ESG) are the current yardsticks of corporate sustainability, assessing the intrinsic and social values of companies from three major aspects. In terms of ecological environment, ESG can increase the business awareness on the treatment of exhaust pollutants. As for social development, ESG can promote a balanced labor market, protect employees' rights and interests. Regards to economic activities, ESG can promote the global industrial chain to pay more attention to environmental and social responsibility and strengthen the efficiency of resource allocation in the capital market. In the post-epidemic era, ESG plays an increasingly significant role in overall economic development. More and more companies and organizations are incorporating ESG mindsets into their business in order to embrace a sustainable future.

The ESG feedback mechanism is an important system to balance the pursuit of business development and sustainability by enhancing market participants' attention to ESG-related factors. The ESG data within the system originates from companies and follows a certain transmission path to capital and resource allocators. ESG disclosure is the first segment in the long chain of data flow, which plays a significant role in transforming unstructured data into structured. Numerous non-profit organizations and government affiliates have been involved in the development of ESG disclosure standards. For example, the Global Sustainability Standards Board (GSSB) has developed the GRI Standards, the non-profit Sustainability Accounting Standards Board (SASB) has developed the SASB Standards, and the non-profit Carbon Disclosure Project (CDP) has developed the CDSB Standards. Then, ESG data providers, companies and organizations, choose the best-applicable or the most cost-effective ESG disclosure standards to output their ESG reports or announcements, under consideration of corresponding industry characteristics, operation model and governance system and regulatory requirements. The structured ESG reports are further interpreted and processed by ESG rating agencies, and transformed into the form of scores. Both the rating scores and ESG reports can better reflect the ESG performance of a company and influence the capital market's evaluation and expectation, the results of which are reflected in financing loan rates and stock market returns. Better ESG performance can lead to favorable loan interest rates, and also convey to investors the company's bright development direction and potentially in-

crease the company's market value. The benefits of the capital market offer positive feedback to the disclosing entity and are also recognition and incentive for it to invest more in the improvement of ESG, which is the essence and starting point of the ESG concept. Therefore, the smooth operation of the feedback mechanism not only gives companies more practical benefits and incentives, transmits effective information to investors and helps them make optimal investment decisions, but also indeed feeds ESG itself, driving both business value and social impacts.

However, the ESG feedback system is not performing properly as things stand. Firstly, the diversity and selectivity of ESG disclosure standards directly lead to different core indicators covered in the ESG reports, which represent the decreasing comparability, and further cause divergence of rating results and investment decisions, greatly weakening companies' motivation for disclosure. Moreover, the phenomenon of greenwashing curtails the effectiveness of the feedback mechanism. Some companies underreport their ESG impacts and circumvent certain regulations, in order to whitewash their ESG performances. They may also receive more favorable interest rate offers, which is a behavior that seriously undermines market equity. To curb the occurrence of unfair greenwashing, it is necessary to strictly control the source of ESG information disclosure at the root. If the regulators mandate all companies to disclose sustainability information according to an extremely comprehensive and detailed disclosure standard, impractical in reality though, it may effectively prevent companies from taking advantage of the flexibility of the standards, thus ensuring equity in the market and enabling the positive functioning of the ESG feedback system. Nevertheless, the sudden increase in disclosure costs and low disclosure efficiency is associated with the critical change of ESG disclosure standards, and some companies may choose to give up disclosing ESG data after weighing the costs and benefits, further hindering the promotion of ESG.

Therefore, we believe that the ESG disclosure standards are the standpoint and foundation of the ESG feedback mechanism, which deeply affects its effectiveness. It can also be regarded as the system of the ESG market, determining the core issues stakeholders should care about and the direction of the overall development. Since the game of equity and efficiency is a constant topic in system design. We find that the trade-off between the two is also reflected in the ESG disclosure standards and further reflected in different channels of the ESG feedback system. Among the current mainstream ESG standards, GRI and SASB can be considered

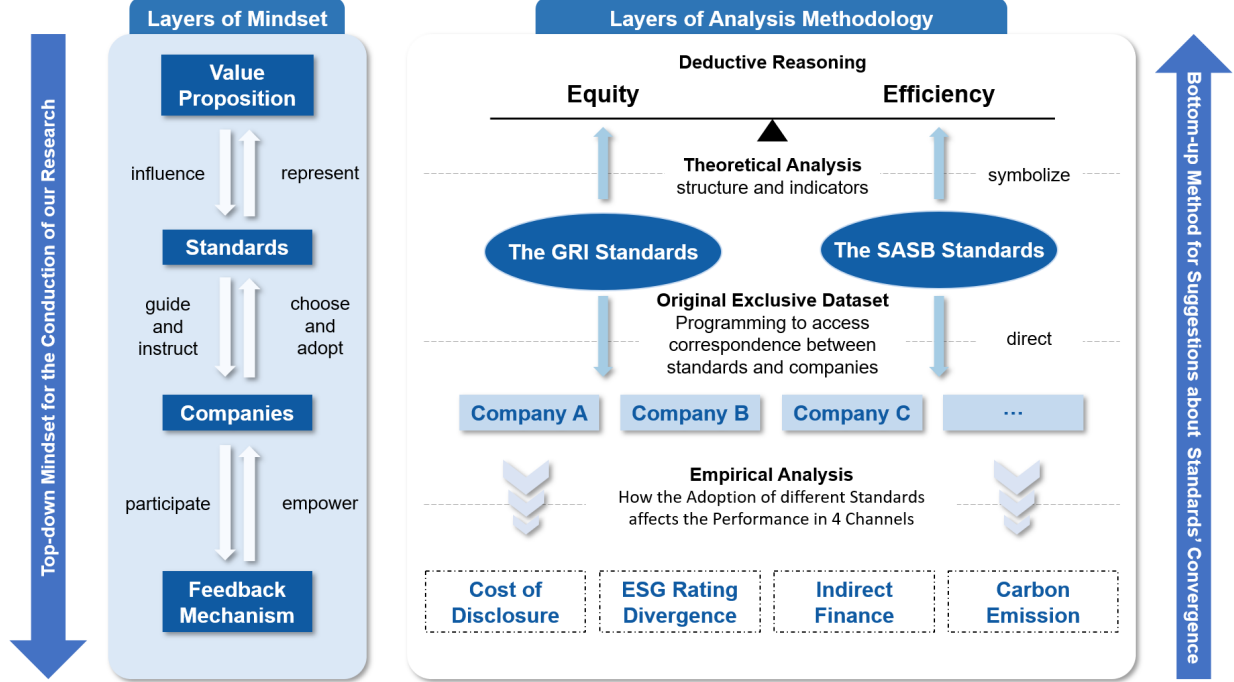
the best representatives of equity and efficiency, respectively. GRI offers ESG data providers a wider range of choices, allowing them to make their own disclosure decisions according to their interpretation, while SASB requires them to follow stringent industry-focused disclosure standards.

The continuous tradeoff between equity and efficiency will not terminate, while reaching an effective balance has also become a common pursuit. In the aspect of ESG disclosure standards, we observe the promising trend of alignment and convergence among various standards, driven by leading initiatives global wide. GRI and SASB, two of the most influential ESG disclosure standards in the world, are critical participants in the process of alignment, concretely representing the symbiotic relationship between equity and efficiency.

In this paper, we propose an original approach for analyzing the difference between two ESG data disclosure standards (GRI and SASB Standards). We first construct a firsthand database from textual analysis from firms' ESG reports, which makes it available for us to use standard data at the firm level to do several empirical tests. From theoretical perspective, we unearth the values hidden behind the standards by analyzing the framework and important issues of standard documents. We deduce that the GRI standard can represent the concept of "equity" and the SASB of "efficiency". We link the standards with the hotly debated values of the ESG system, and give directions for the future convergence of "ESG standard zoo", which could be considered similar as the "factor zoo" but in the field of ESG. And more importantly, we give empirical evidence to support the necessity and direction of standard convergence. From empirical perspective, we design tests through four channels (ESG data disclosure, ESG rating and direct finance, indirect finance, improvement of ESG itself) of ESG feedback system. We find that firms are motivated to actively choose ESG disclosure standards through disclosure costs, while large firms are more inclined to use the GRI standard to disclose more quantitative data and growing firms prefer the SASB. An interesting point is that although the use of GRI promotes corporate emission reduction, banks are often more willing to refer to information under the SASB Standards when considering loan interest rates than the GRI's. One of our core marginal contributions lies in the impact of ESG data disclosure standards on direct corporate finance. We use a moderating effect model that confirms GRI's assistance to weaken the ESG rating divergence of the same individual company, and further discover GRI's improvement on differentiation of separate companies,

which significantly reduces the confusion of direct investors. In this approach, we suggest that the efficiency-focused SASB may lack efficiency in the field of direct finance.

Figure 1: Research Framework



Note: This figure shows the framework of our paper in terms of general mindset and analysis methodology. Our research idea can be divided into four layers: value, standards, companies and the feedback mechanism. On the one hand, the value proposition of the standard setter would influence the structure and indicators of the standard itself. Companies would disclose and manage their ESG data based on the guidance and instruction provided by the certain ESG disclosure standards they chose. Meanwhile, companies are fundamental participants in the ESG feedback mechanism. On the other hand, the feedback mechanism would empower and incentivize companies to improve their ESG performances. The standards can also represent its embedded core value. As for specific analysis, we discover that the GRI Standards and the SASB Standards are the best representatives of equity and efficiency respectively, which indicates a classic game between the two. This inference is concluded after our deductive reasoning and theoretical analysis based on former sustainability theories and the structure and indicators of the two most popular-used standards. Moreover, after programming to access correspondence between standards and companies, we conduct empirical analysis and reveal how the adoption of different standards might affect companies' performance in 4 channels, which are ESG data disclosure, ESG rating and direct finance, indirect finance and improvement of ESG itself. The cost of disclosure, ESG rating divergence, indirect finance (basically loan interest rates) and carbon emission are the sub-channels we focused on, and are major concerns related to the 4 channels respectively. Based on comprehensive analysis and detailed reasoning, we finally form a set of suggestions about the convergence of ESG disclosure standards.

In order to further empirically test the significant difference between GRI and SASB in

equity and efficiency, we derive four assumptions from materiality and flexibility, and carry out strict tests one by one in the subsequent empirical analysis part. First of all, we believe that the GRI, which adopts double materiality, is likely to put forward more comprehensive disclosure requirements for companies (Machado et al., 2021), resulting in higher disclosure costs. On the contrary, the SASB, which holds single materiality, will only put forward higher disclosure requirements for companies in financial aspects, resulting in lower disclosure costs and higher efficiency. In the empirical analysis, we find that the ESG reports using GRI for disclosure show significantly more document pages and more data points, and small growth companies with high leverage and high Tobin Q are more likely to use SASB with lower disclosure costs. These findings are strong arguments for our first hypothesis.

Secondly, we believe that the GRI's focus on dual substantive values is more consistent with ESG rating agencies. Companies that use the GRI Standards to disclose should be able to convey more effective information to rating agencies, so as to bridge the differences between different rating agencies on the same company's ratings, thus improving the decision-making efficiency of direct investors (Plumlee et al., 2015); while the SASB may expand the scoring differences between different rating agencies due to the lack of environmental, social and other information. In the subsequent empirical tests, we construct a moderating effect model, which not only verifies this hypothesis, but also finds that the adoption of GRI will help to reflect the differences between different companies in ESG scores, thus better helping direct investors make decisions; it's worth noting that SASB's failing to do this also reflects its lack of efficiency in direct corporate financing, which is an important conclusion of our research.

Thirdly, we believe that when making decisions to lend to companies, banks coincide with the values of the SASB, which focuses on efficiency. Companies using SASB are likely to obtain lower loan interest rates because of more intensive effective information in the eyes of bankers when holding other conditions constant (Berger et al., 2005). However, the GRI Standards that focus on equity will even show the suspicion that the managers are "showing off" to the banks (Bénabou & Tirole, 2010), so as to improve the financing constraints of companies. This assumption is also verified by the difference in difference (DID) model later.

Finally, we believe that the GRI is likely to urge the company to disclose ESG information in a scientific way, and at the same time promote its positive transformation in aspects such as environmental protection and social responsibility (Cong et al., 2022), while SASB lacks

this function. The subsequent empirical test also verifies the assumption that companies prefer green production mode after using the GRI standard through indicators such as carbon emissions.

Our study is related to a rapidly growing literature on ESG. Firstly, there is a growing number of academic studies which rely on ESG ratings for their empirical analysis (Servaes & Tamayo, 2013; Flammer, 2015; Liang & Renneboog, 2017; Lins et al., 2017; Albuquerque et al., 2019). As a result, ESG ratings increasingly influence decisions, with potentially far-reaching effects on asset prices and corporate policies. Secondly, We extend existing research that has investigated the divergence of ESG ratings (Chatterji et al., 2016; Gibson Brandon et al., 2021; Christensen et al., 2022; Berg et al., 2022). The aim of ESG data is to measure and assess the performance of corporations on a large set of issues, which are often categorized according to the environment, social and governance trichotomy. The importance of this data can hardly be overstated, as it drives the decisions of all SR investors around the world. However, with abundance and variety comes heterogeneity: it is common that rating agencies to disagree in their conclusions. This may cause headaches, as a strategy’s returns can very well change when switching from one data provider to another. Disagreement is well documented not only for firms, but also for funds (Gangi et al., 2022). However, there is still little research on ESG data disclosure standards, especially using empirical analysis at the company level. Therefore, we use the first-hand data of the company’s ESG report as a channel to analyze the more far-reaching impact of the company’s adoption of ESG disclosure standards.

The remainder of the paper proceeds as follows. Section 2 provides a conceptual analysis to characterize ESG disclosure standards and develop four hypotheses of how ESG data disclosure has an influence on ESG systems. In Section 3, we describe the data and provide summary statistics. We discuss the results of empirical analysis in Section 4. Concluding remarks are in Section 5.

2 Conceptual Framework

2.1 Theoretical Backgrounds

The historical process of the industrial revolution has accelerated the economic development of human beings, but the ensuing negative problems such as natural resource shortage and

environmental pollution have also triggered profound reflections. After being nourished by externalities theory, corporate social responsibility theory, and sustainable development theory, ESG theory has become a synthesis of these three, promoting the further balance between economic development and external impacts, as well as the game between efficiency and equity.

Externalities theory developed from Principles of Economics ([Marshall, 1890](#)), The Economics of Welfare ([Pigou, 2018](#)), and The Problem of Social Cost ([Coase, 1960](#)), and it is concerned that ecological and environmental resources, as a kind of public good with unclear property rights, cannot be solved entirely by market mechanisms. Instead, government intervention and control are needed. Therefore, externality theory focuses on the balance between environmental impact and business development, symbolizing the game of equity and efficiency respectively, which is also reflected in the Environment part of the current ESG theory.

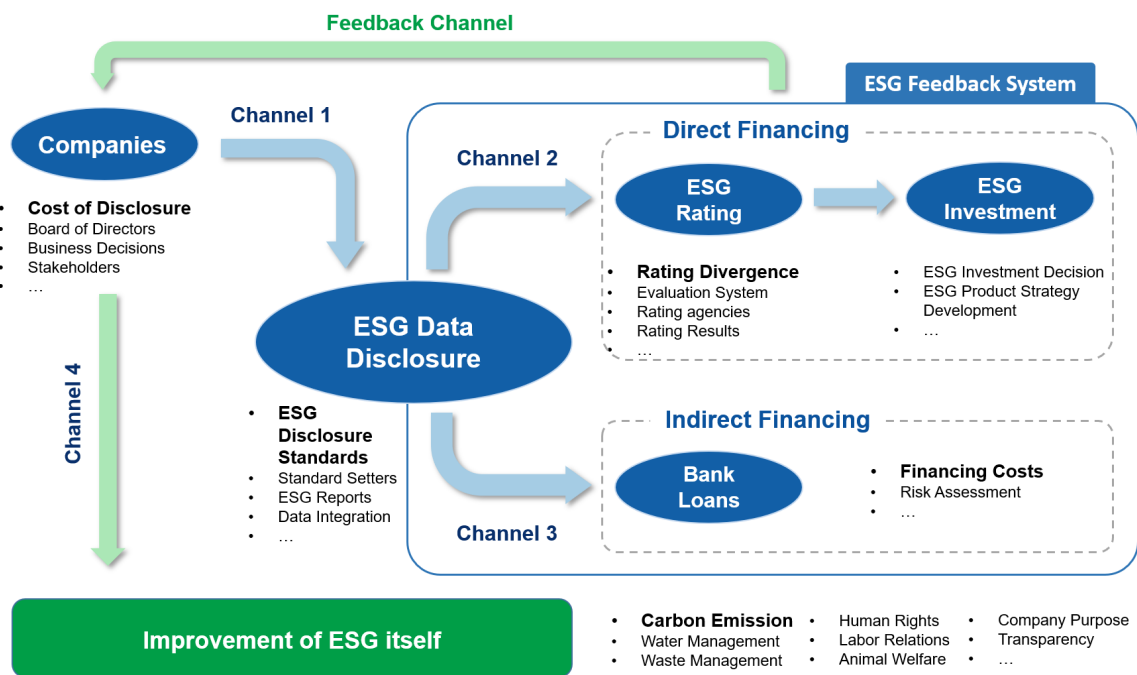
Corporate social responsibility theory emerged in the 1960s, and originated from two schools, shareholder supremacy and stakeholder doctrine. However, stakeholder doctrine began to rise in the 1980s with the publication of theories by R. Edward Freeman, Archie B. Carroll, and John Elkington, which emphasized that corporate development should balance financial returns and social benefits. Thus, CSR theory focuses on the game of social impact, corporate governance dimensions and corporate financial development, reflecting equally equity and efficiency, as reflected in the S and G parts of the current ESG theory, and, of course, with some influence on E.

The theory of sustainable development originated in 1962 with the publication of the environmental science book *Silent Spring* by the American marine biologist Rachel Carlson ([Rachel et al., 1962](#)). At that time, people continued to reflect on the environmental and social problems brought about by the Anthropocentrism mindset, which prompted legislatures and regulatory agencies to intervene in the environmental externalities generated by business activities and gave rise to the Ecocentrism mindset. After about 18 years of collision between these two paradigms, the concept of sustainable development was developed by the International Union for Conservation of Nature and Natural Resources (IUCN) in its World Conservation Strategy published in 1980 and matured by the United Nations in 1987 with the release of *Our Common Future* ([UNEP, 1980](#)). In Chapter 2 of *Our Common Future*, sustainable development is defined as “development that meets the needs of the present without compromising the ability of future generations to meet their needs” ([WCED, 1987](#)), which

directly reflects the concept of intergenerational equity and inclusive development, which also provides important inspiration for the foundation of ESG theory.

Nowadays, ESG theory is increasingly concerned by business and capital providers, which not only focuses on externality, corporate social responsibility and intergenerational equity, but also integrates intragenerational equity, emphasizing the balance of economic development and ESG in both current and future time dimensions. It further improves the system of checks and balances between equity and efficiency, and promotes the long-term value of both companies and macro economies.

Figure 2: The ESG Feedback and Output System



Note: This figure shows the framework of ESG feedback and output system, which consists of five main components, companies, direct finance, indirect finance, improvement of ESG itself and ESG data disclosure. The first four components are all linked surrounding the last one with four channels. The word “feedback” means the incentive mechanism when companies receive more benefits if they have better ESG performances. The word “output” means the companies’ improvement of their ESG performances, represents not only the increasing current value of companies but also their sustainability.

ESG theory has now developed a relatively clear operating system in the business and capital market, lacking a well-established positive feedback mechanism though. On the whole, ESG information originates from enterprises, follows a certain transmission path, and eventually reaches resource allocators and feeds ESG itself. The units involved contains ESG data

providers (corporations and organizations), ESG disclosure standards, ESG rating frameworks, indirect investment providers, direct investment providers and ESG itself. The relationships between different entities could be concluded as 4 channels, which are, respectively, ESG data disclosure (Channel 1), ESG rating and direct finance (Channel 2), indirect finance (Channel 3) and improvement of ESG itself (Channel 4). Additionally, channel 1 could be identified as the key process of the ESG system from the perspective of corporations and organizations, while channels 2 and 3 could be concluded as from the perspective of investors, channel 4 could be interpreted as from the perspective of broader stakeholders.

ESG data disclosure is undoubtedly a critical component in the operation of the above channels, directly determining the feedback on channels 2 and 3, and contributing to the implementation of channel 4 with the disclosure of ESG information by companies and organizations themselves. Firstly, ESG information of companies and organizations is transformed from unstructured data to structured data, which is reflected in Channel 1, ESG data disclosure. Secondly, ESG disclosure standards significantly enhance the comparability and quantification of ESG data among different companies and promote the efficiency of investments in channel 2 and channel 3. For Channel 2, ESG rating and direct finance, ESG data disclosed by companies is first interpreted by ESG rating agencies and then transformed into their ESG rating scores and expressed in the capital market, so that direct investors may make investment decisions with a certain degree of reference to the scores. For Channel 3, indirect finance, banks and other indirect investment institutions usually refer to the ESG report issued by the company, evaluate its ESG performance and conduct a comparative analysis, and then make decisions such as reducing the lending rate. Thirdly, the values and core indicators embedded in the ESG disclosure standards also provide guidance for organizations to enhance their ESG performance, increasing the efficiency of channel 4, and improvement of ESG itself.

Table 1: Comparison of main ESG Disclosure Standards

Panel A: Basic Information							
Standards	Standard-setting Organization	Reference Ratio	Year of the First Version	Year of the Latest Version	Count of Revision	Target Audience	Scope of Materiality
GRI	GRI	95%	2000	2020	4	Multiple Stakeholders	Financial, Environmental, Social
SASB	SASB	78%	2018	2018	0	Capital Providers	Financial
IIRC	IIRC	75%	2013	2021	1	Capital Providers	Financial
CDP	CDP	67%	2003	N.A.	N.A.	Multiple Stakeholders	Financial, Environmental, Social
TCFD	FSB	57%	2017	2017	0	Capital Providers	Financial
CDSB	CDSB	30%	2015	2018	1	Capital Providers	Financial
ISSB	ISSB	N.A.	N.A.	N.A.	N.A.	Capital Providers	Financial

Panel B: Range, Applicability and Goal			
Standards	Scope of Disclosure Topics	Cross-industry Applicability	Main Purpose
GRI	Economy, Environment, Social, Governance	General and Topic standards are cross-industries applicable, while industry standards under developed	Set criteria for ESG-related indicators
SASB	Environment, Social, Governance	Industry Featured	Disclose financially significant sustainable development information
IIRC	Economy, Environment, Social, Governance	Cross-industry applicable	Reveal financial and non-financial correlations to illustrate long-term competitiveness
CDP	Environment	Cross-industry applicable, complemented industry-specific contents	Make companies and cities aware of the environmental impact
TCFD	Climate-specific	Cross-industry applicable, complemented industry-specific contents	Disclose information on risks/opportunities related to climate change
CDSB	Environment	Cross-industry applicable	Integrate environmental information into financial information
ISSB	General disclosure (environmental, social, governance) and climate specific	General disclosures is cross-industries applicable, while climate related disclosures is industry featured	N.A.

Panel C: Structure and Topics

Standards	Structural Framework	Core Issues
GRI	General, Industries, Topics (2020 version)	Economic, Environmental and Social
SASB	77 Industries from the perspective of 6 Sections	Environment, Social Capital, Human Capital, Business Models and Innovation, Leadership and Governance
IIRC	Guiding Principles, Content Elements	External Environment, Governance, Business Model, Risks and Opportunities, Strategy and Resource Allocation, Performance
CDP	3 Independent Modules	Climate, Forests, Water Security
TCFD	4 Main Sections	Governance, Strategy, Risk management, Metrics and Targets
CDSB	6 Environmental Modules	Climate Risks and Opportunities
ISSB	N.A.	N.A.

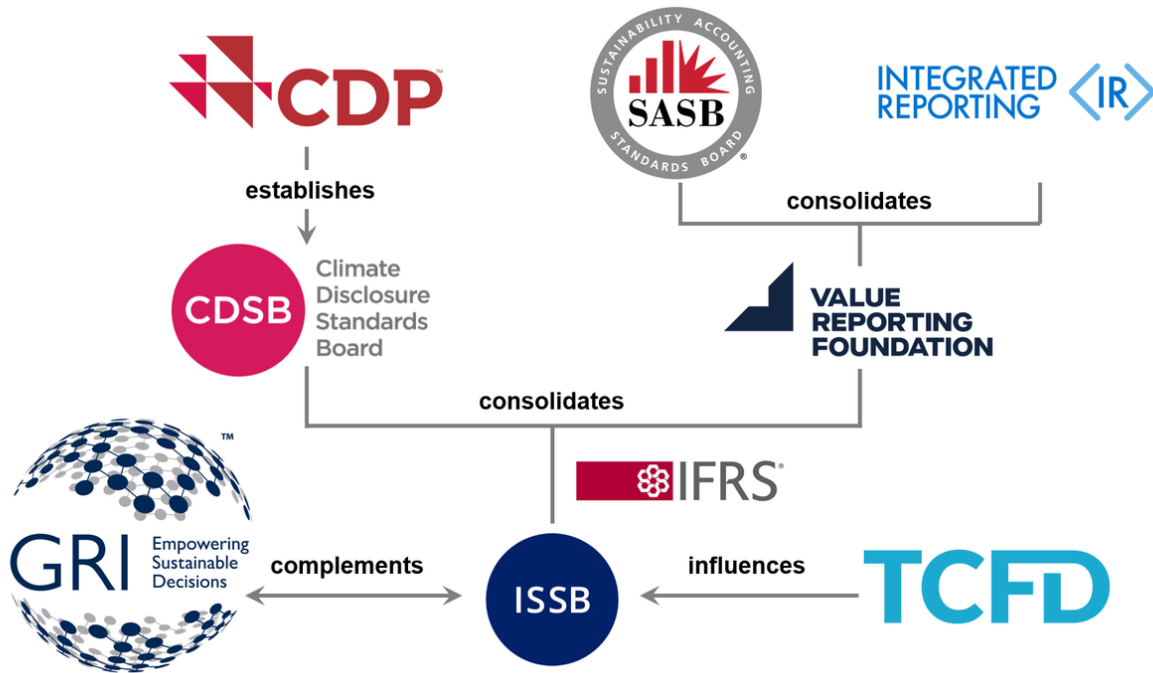
Note: This table shows the characteristics of seven main ESG disclosure standards' structure and topics.

A growing number of players play a variety of roles in the sustainability information arena, including investors, businesses, regulators, nonprofits, and other civil society organizations. Together, they form a dynamic ecosystem of sustainability information, and ESG disclosure standards are the foundation of the ESG standards ecosystem. Standard setters, consisting of nonprofit organizations or government affiliates, develop standards that incorporate their own understanding of ESG and market demand for ESG, identifying elements such as the target audience, scope of disclosure information, and cross-industry applicability of the standards. The most influential international ESG disclosure standards currently include Global Reporting Initiative (GRI), Sustainability Accounting Standards Board (SASB), Task Force on Climate-related Financial Disclosures (TCFD), Carbon Disclosure Project (CDP), Climate Disclosure Standards Board (CDSB), International Integrated Reporting Council (IIRC) and International Sustainability Standards Board (ISSB). The characteristics of existing ESG standards vary due to the different understanding of ESG by standard setters and the different

focus of standard development. The contents of the standards complement each other and are applicable to different types of companies and organizations. With the exception of SASB, which is industry-specific, several of the major ESG standards are currently applied across industries. According to the official reports of each standard-setting body and rating agency, in terms of materiality, the GRI and CDP focus on financial, environmental, social and other comprehensive information, while the remaining five standards represented by SASB are all focused on financial materiality. In terms of information disclosure scope, CDP, CDSB and TCFD only disclose environmental information, while the rest of the standards represented by GRI disclose comprehensive information including environmental, social and economic aspects.

In July 2020, GRI and SASB announced a joint project to develop and promote a mutual exploration of company experiences in using both sets of standards, and in April 2021, they jointly published a practical guide to sustainability reporting. The SASB issued sustainability accounting standards in 2018, focusing on financial material information and providing service for capital providers. The IIRC issued a revised integrated reporting standard in January 2021, linking financial information to non-financial information to reflect and explain the long-term competitiveness of companies. To simplify the corporate reporting system, provide investors and businesses with an integrated corporate reporting standard covering all corporate values, and continue to promote global sustainability, in June 2021 IIRC and SASB formally announced their merger to create the Value Reporting Foundation (VRF). In November of the same year, the IFRS Foundation established the new International Sustainability Standards Board (ISSB), a standard for reporting environmental and climate change information for businesses and national governments, created by the non-profit organization CDP to provide companies with a standard for reporting environmental and social information that is as rigorous as their financial information. With many investors and regulators calling for globally comparable sustainability financial reporting and looking to incorporate IFRS expertise, the CDSB and VRF were merged together into the IFRS Foundation in 2022. The merger of the CDSB and VRF marks the simplification and standardization of corporate sustainability disclosure and sets the stage for the establishment of a global sustainability disclosure standard. The ISSB will therefore blend the value context of the CDSB and VRF to develop a set of ISSB disclosure standards that enhance harmonization and focus on financial information relevant to impacts on sustainability.

Figure 3: The Trend of Alignment among Different ESG Disclosure Standards



Note: This figure shows the process and logic of alignment among main ESG disclosure standards.

Unlike SASB, GRI, a set of globally widely used and diverse disclosure standards, focuses on integrated environmental, social and financial information and aims to disclose its environmental and social impacts to a wider range of stakeholders than investors. And the two standards, which have very different information coverage and cross-industry application, signed a memorandum of understanding in 2022 announcing that they would work together to develop an international standard for sustainable disclosure. The two decided to divide the standard into two parts, capital user-focused and multi-stakeholder-focused, with the GRI standard setter and SASB taking responsibility and agreeing on the standard separately. This shows that GRI and SASB are the two fundamental core sources for the convergence of ESG standards, representing the convergence trend of multiple interests and single capital, and leading the cooperative convergence of the two mainstream standards. The process of revising the GRI 2016 to 2020 edition also learns from the industry classification characteristics of SASB, taking industry standards into consideration and linking them to topic standards to enhance the efficiency of disclosure. The revision and convergence of standards fully reflect the combined efforts of the standards for fairness and efficiency.

2.2 Foundation: Equity or efficiency? Evidence from GRI and SASB Standards

2.2.1 Deductive Reasoning

For any set of standards, its composition generally consists of two aspects, structure and indicators. The structure refers to the arrangement of the content of the standards, while the indicators are the specific embodiment of the content of the standards. For ESG disclosure standards, in addition to the basic characteristics of the structure and indicators, there are also their respective attributes. Given that we have mainly focused on the GRI and SASB standards, we have extracted two dimensions of attributes based on them, materiality and flexibility.

The legal notion of “materiality” serves as the cornerstone of the US corporate disclosure standards. It offers the theoretical framework for the disclosure of specific data that investors use to cast ballots and decide which investments to make. Materiality in ESG disclosure standards refers to the significance or relevance of specific ESG issues for a particular company or industry, and it could be further identified as single materiality or double materiality. Single materiality is a reporting approach that takes into account how sustainable elements affect a company’s financial performance, whereas double materiality also considers the company’s impact on the environment and society as a whole. Both of these could be considered as the game between shareholder supremacy and stakeholder doctrine, and determine the value proposition of the entire standard, reflected in the structure and topics of the standard.

Flexibility could be understood from different aspects. Particularly in the context of ESG disclosure standards, the concept of “flexibility” could be interpreted from the compulsory level of disclosure, the quantitative level of the standard requirements and the scope of disclosure indicator alternatives. Since quantitative data is more comparable and costs more for companies to determine the absolute value, qualitative data relies more on companies’ self-declaration and has more space for them to manipulate, we conclude that when a standard requires a company or organization to disclose more qualitative data compared to quantitative data, the standard could be considered more flexible. When a company could choose whether to disclose a specific indicator on a voluntary basis but not on a compulsory basis or has a larger number of alternatives for choosing which specific indicator to disclose, the correspond-

ing standard could be considered more flexible since it offers more free choices for the ESG data providers.

The relationship between materiality and the game of equity or efficiency is clear. Standards based on double materiality propositions could be considered to represent a larger degree of equity, since it not only takes business development into account, but also cares about other stakeholders, society and the environment as a whole, which implies intragenerational equity and intergenerational equity based on the value of sustainable development. Although the relationship between flexibility and the game of equity or efficiency remains quite vague, when it comes to the context of specific ESG disclosure standards, we could deduce that when a standard is comparatively flexible under the direction of double materiality, the representativeness of equity is amplified. From the perspective of companies and organizations, they have the right and opportunity to choose the best-matched or the most cost-effective indicator under the guidance of double materiality, which implies freedom of choice, a significant component of the equity theory. Without certain regulations and laws, the flexibility of disclosure may increase the ESG data providers' incentive to disclose and further the implementation of double materiality, which is beneficial for both the business and ESG fields. However, the more flexible the disclosure is, the more risks of non-comparable there are. With the decreasing comparability, rating agencies and investors would spend more effort to formulate their decisions, which decreases the efficiency of the ESG system functioning. Besides, the freedom of choice implies increasing decision-making costs and the risk of choosing non-core indicators to disclose intentionally and further reducing the efficiency of both data providers and data users.

To conclude, when an ESG disclosure standard demonstrates the characteristics of double materiality and flexibility simultaneously, it could be categorized as a standard more influenced by the concept of equity rather than efficiency. Meanwhile, with single or financial materiality and strictness, a standard could be seen as a representative of efficiency. We must admit this categorization methodology is not accurate, but the logic and value behind it are rigorous and reasonable, which implies the game of equity and efficiency with regard to the field of ESG disclosure standards.

2.2.2 Materiality

GRI and SASB focus on different dimensions of materiality: the GRI is a set of ESG standards with double materiality, while the SASB is more concerned with financial materiality. The GRI Standards is based on the principles of balance, comparability, integrity, and verifiability, and helps companies understand the impact of their business activities on sustainability issues. Under the GRI Standards, companies disclose information to multiple stakeholders, including the environment, society, investors, and the public. SASB is a set of sustainability accounting standards covering 77 industries, which aims to help companies communicate financially significant sustainability-related information to investors, so that they can make more scientific and rational investment decisions. The information disclosed by companies under the SASB Standards is mainly for the economic stakeholders, such as the debt and equity of the company, thus SASB focuses more on the disclosure of ESG information related to finance.

Table 2: Scores of GRI and SASB Standards from Section Perspective

Section	Total		GRI 2020		SASB 2018	
	Number of Index	Score	Absolute Score	Final Score (100 in total)	Absolute Score	Final Score (100 in total)
Environment	22	66	56	84.8	49	74.2
Social	18	54	37	68.5	39	72.2
Governance	33	99	86	86.9	43	43.4
Strategy	27	81	48	59.3	61	75.3

Note: This table shows the total index number and score, the absolute score and the percentage version of GRI and SASB standards from the perspective of four sections, including Environment, Social, Governance and Strategy. GRI and SASB standards' versions are respectively 2020 and 2018.

In terms of indicators, we rated their coverage in the four sections of Environment, Social, Governance, and Strategy, and compared the results on a percentage scale. Firstly, by combining the Environment, Social and Governance aspects of ESG and the Strategy section in the TCFD financial disclosure standard, we have divided the above four first-level sections. Then, we split the four sections into 22, 18, 33, and 27 sub-sections respectively to further the measurement of ESG standards' materiality. For example, the environment section includes GHG emissions, water management; the social section includes employee engagement, human rights;

the government section includes company purpose, board diversity, equity and inclusion; and the strategy section includes revenue pipeline, capital allocation and other financial-related dimensions. Secondly, based on our understanding of the GRI and SASB standards and online interviews with the ESG Navigator* management team, as well as a combination of ESG data providers, ESG standard makers, and ESG rating agencies' official disclosures and declarations, we rated the level of GRI and SASB concern in each section. Each section has three levels of scores, 1, 2, and 3. The higher the score, the higher the coverage of the standard in this section, i.e., the more emphasis is placed on it. Thirdly, after calculating the total scores of GRI and SASB in Environment, Social, Governance, and Strategy (66, 54, 99, and 81, respectively), we normalized them by dividing the scores obtained by GRI and SASB in each section by the total score of that section and multiplying by 100 to obtain the comparable score of the two standards in each section.

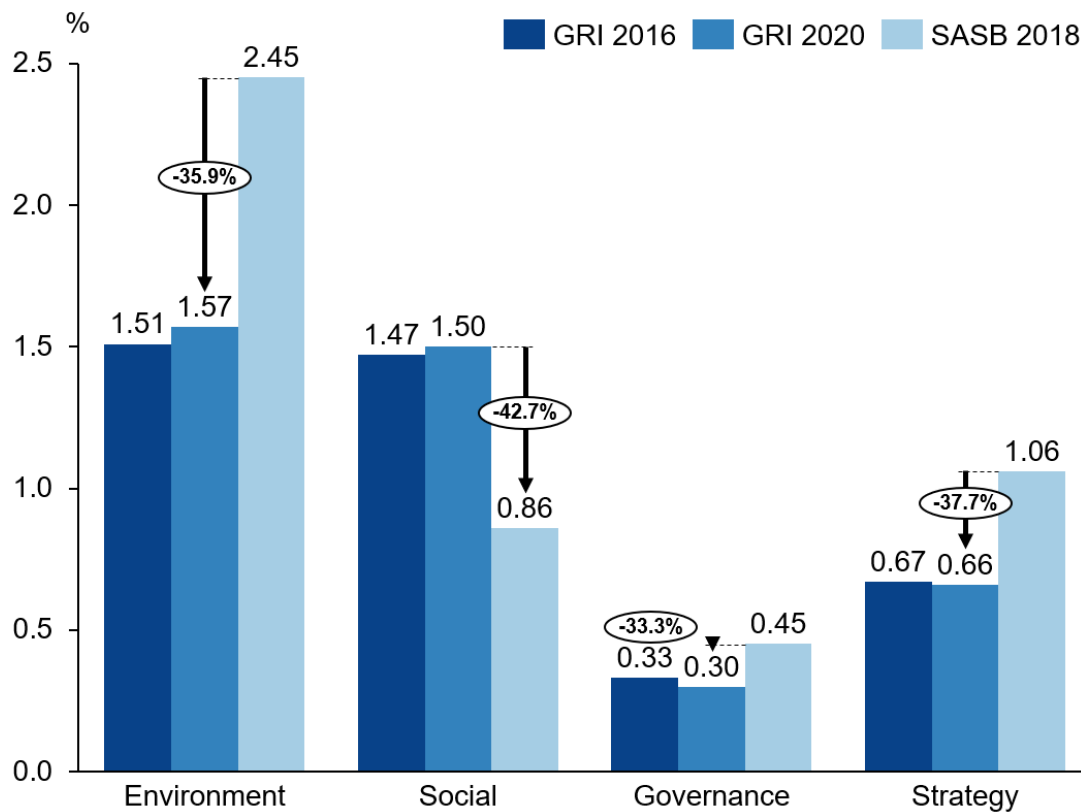
The GRI scored 299.5 overall, higher than the SASB total score of 265.2. This partly confirms the double materiality attribute of the GRI standard, as it focuses on a wider variety of aspects. Specifically, in terms of environment and governance, GRI scores significantly higher than SASB, with a difference of 10.6 and 43.5 points respectively; and in terms of strategy, SASB scores significantly higher than GRI, with a difference of 16.0 points; while the difference between the two scores was not significant in the social section.

GRI's leading score in the environmental section and our judgment of its double materiality show a clear consistency, while SASB's high score in Strategy also confirms its financial materiality. As for the governance section, SASB has publicly stated that it deliberately avoids some governance-related indicators to reduce repetitive information, since it believes those indicators are already published in companies' annual reports. This claim shows that SASB helps companies disclose ESG information efficiently and emphasizes cost-effective financial materiality, which further reflects SASB's financial materiality.

In addition to the inductive statistics, we also conducted a textual analysis of GRI2016, GRI2020 and SASB2018 standards. We first summarize the words related to the above four sections (see Appendix 7.3, Word Bank by Sections), then write a program using Python to count the word frequency share, and finally sum up the words according to the sections they belonged to quantitatively verify the difference of materiality between the two standards.

*<https://esgnavigator.com/>

Figure 4: Word Frequency of GRI and SASB Standards from Section Perspective



Note: The bar chart shows the word frequency related to different sections in GRI and SASB Standards.

Since SASB has developed the disclosure structure by industry, we have directly combined the documents of 77 industries here for word frequency statistics, causing its word frequency slightly over-represented, but the comparison results of the following sections are still valid and credible.

The distribution of detailed indicators under the GRI standard before and after the revision remains basically the same in the four sections. Comparing GRI2020 and SASB2018, we discover that in the social section, the GRI word frequency is nearly twice of SASB (1.5% and 0.86% respectively). Excluding the influence of SASB in the summation analysis that leads to over-represented data, the difference in the social section will be even more significant. This is reflected in the fact that GRI pays more attention to social conditions such as employee safety and the surrounding communities. In the strategy section, the relevant words frequency of GRI is significantly lower than that of SASB (0.66% and 1.06% respectively), which reveals that SASB standards have more content related to corporate financial information such as

economic, financial, and value. This is consistent with the conclusion that GRI focuses on double materiality while SASB focuses on financial materiality.

Based on the scoring and word frequency analysis of the four sections, the emphasis of GRI and SASB standards is demonstrated, and the double-materiality of GRI and the financial materiality of SASB are confirmed. In regard to ESG disclosure standards, double-materiality represents a company's focus on comprehensive performance, giving sufficient attention to environmental, social, and corporate governance aspects, and incorporating different influencing factors into the scope of ESG report assessment. The comprehensive disclosure is valuable for multiple stakeholders, meeting their needs and facilitating the formation of a more complete view. It can also better reflect the overall performance of the company's business activities, which might be conducive to maintaining long-term fairness and equity in the market and providing a dynamic environment for investors to make decisions. Financial-materiality, on the other hand, means that companies focus only on ESG disclosure that is closely related to its business and finance, weakening the disclosure comprehensiveness. Some investors may be influenced by short-sightedness and drive other irrational investors to make biased judgments, sacrificing the fairness and equity of the business environment to a certain extent. However, since the standard adopted financial materiality focuses on a single piece of financial information, which significantly reduces the amount of information that needs to be taken into account in the process of ESG disclosure, and further increases efficiency at the executive level.

2.2.3 Flexibility

As a whole, the GRI Standards exhibits more pronounced flexibility than the SASB Standards, which is reflected in its structure and indicators. As in the deduction above, the indicators can be drawn from the compulsory level of disclosure, the quantitative level of the standard requirements and the scope of disclosure indicator alternatives, three dimensions of flexibility.

At the structural level, GRI is generalized and modularized, while SASB is industry-focused and within-modularized. (see Appendix 7.4, Comparison between the Structure of GRI and SASB Standards) The GRI Standards, version of 2020, has established three categories of standards: general, industry and topic. Companies in any industry need to start with general standards, and then find and select specific topics in the topic standards that they need or

intend to disclose. Companies in some relevant industries can refer to industry standards for indexing, and industry standards supplement some industry-specific indicators that are not covered in the topic standards. Prior to the revision of GRI in 2020, industry standards were not included in GRI. For any company disclosing using the GRI Standards, it was necessary to refer to all the topic standards and choose specific disclosure items independently, which increased the range of independent choices for companies. Meanwhile, the modularized design of GRI can more easily manage and update various topics. For standard setters, it is possible to flexibly modify the modules without affecting the availability of the entire standard. This greatly simplifies the process of updating and revising. Therefore, as far as the standard itself is concerned, GRI is more variable and allows more flexible editorial processing.

In fact, the structure of the SASB Standards is its self-established industry classification system, commonly known as the Sustainable Industry Classification System® (SICS®). The standards developed into 77 specific industries corresponding to its classification system. Taking all the industry standards as a whole, it includes five sustainability modules defined by SASB, containing 26 specific topics; the specific topics covered in each are selected based on the Materiality Map, which is initiated by SASB based on the development and business characteristics of each industry. The selection process is done by the standard-setter for the enterprise first, and the enterprise only needs to comply with it, representing a certain extent losses of the freedom to choose. The SASB Standards also demonstrate the characteristics of modularization. However, unlike GRI, it is implemented WITHIN-modularized under the premise of differentiating industries. SASB notes that when a company’s main business scope spans multiple industries, companies should combine their disclosures with references to multiple industry standards. The design of the framework makes it more complex for SASB to manage updates on topics, and when macro policy changes occur, SASB needs to make changes to each of the 77 standards. It can be seen that, as far as the standards themselves are concerned, the SASB Standards are more difficult to adapt to and to a certain extent reflect strictness.

At the indicator level, we similarly conducted a textual analysis of GRI and SASB. We initially focus on the quantitative and qualitative levels of specific indicators in the standards. Qualitative indicators refer to disclosure requirements that contain verbs such as “describe”, “explain”, etc. Quantitative indicators refer to disclosure requirements that require the dis-

closure of specific numbers, such as “percentage”, “number”, “ratio” and other nouns. We write a python program to retrieve specific pages of the standard text, extract the disclosure requirements under each topic of GRI and the accounting metrics table in each industry standard of SASB, aggregate the disclosure metrics and then manually distinguish the metric categories and count them.

We find that the proportion of qualitative and quantitative indicators in the 2016 and 2020 versions of the GRI standard remained similar, while the number of quantitative indicators in the SASB standard reaches nearly four times that of qualitative indicators. This shows that SASB attaches great importance to the disclosure of quantitative indicators.

Table 3: Qualitative and Quantitative Indicator Statistics of GRI and SASB

Data	GRI 2016		GRI 2020		SASB 2018	
	Number of Data Points	Proportion (%)	Number of Data Points	Proportion (%)	Number of Data Points	Proportion (%)
Total	540	-	607	-	1107	-
Qualitative	283	52.4	334	55.0	225	20.3
Quantitative	257	47.6	273	45.0	882	79.7

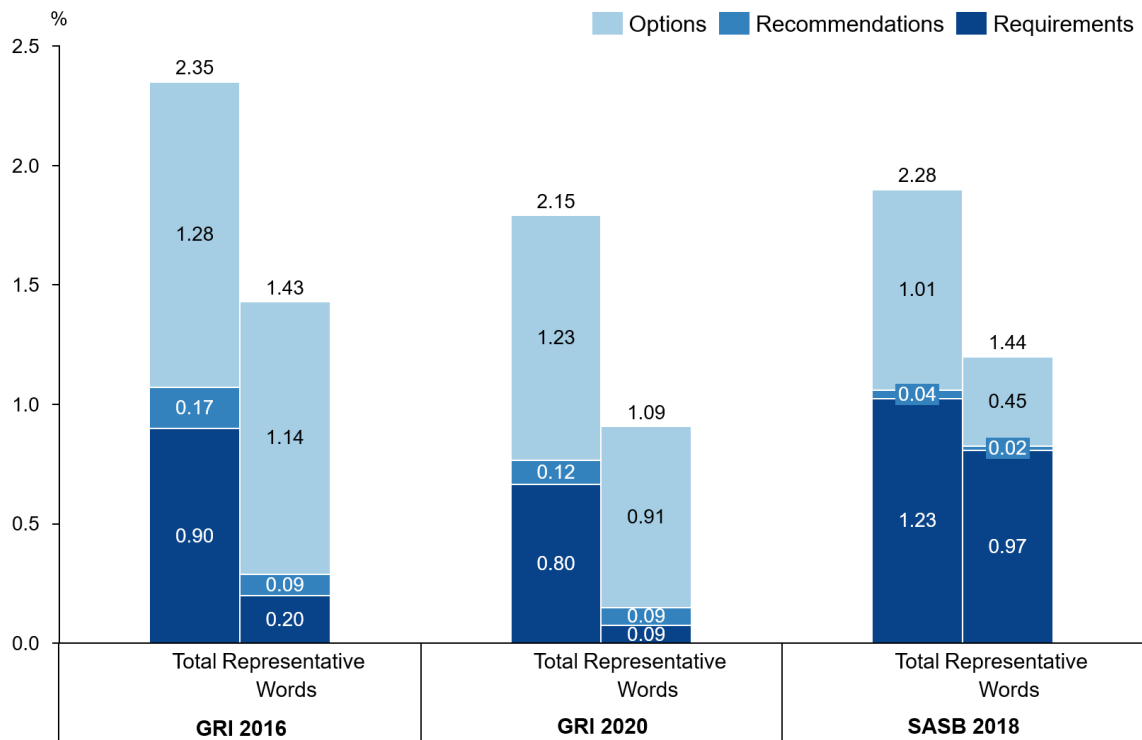
Note: This table shows the absolute number and the percentage of qualitative and quantitative data of GRI and SASB standards. The GRI standards has two versions, 2016 and 2020, while SASB is the version in 2018.

We also conduct a word frequency analysis of the mandatory level of disclosure requirements in a similar way to the previous section. According to the official guidelines of GRI and SASB, both of them use “shall”, “should”, and “can” as the core vocabulary to measure the level of compulsory. The term “must” indicates requirements. An organization is required to abide by the rules in order to submit reports that meet the criteria. guidance and recommendations may also be included in the standards. in some situations, taking a certain action is suggested but not obligatory. The words “should” and “can” denote recommendations and options, respectively. However, we are concerned that in addition to the above three terms, there are other terms that represent the mandatory level of disclosure that appears in the standard. Therefore, we divide the mandatory into three levels, Requirements, Recommendations, and Options, and filtered the words around “shall”, “should” and “can” regarding the corpus to find the relevant words that can represent the same level of mandatory, and then aggregated them to form a thesaurus (see Appendix). After that, we use python to write a

program to count the frequency of words.

We find that the frequency of words indicating options in the GRI Standards is significantly higher than that of words indicating requirements and recommendations, more than four times the frequency of the latter two words combined. In contrast, the SASB standard is dominated by words indicating requirements. This shows that SASB has more stringent disclosure requirements at the indicator level, which are mainly mandatory disclosures, while GRI is more inclined to provide support and assistance to companies in the form of recommendations. This reflects the relatively more flexible nature of GRI and the more stringent nature of SASB at the level of mandatory level of disclosure.

Figure 5: Word Frequency of GRI and SASB Standards from Mandatory Level Perspective



Note: The bar chart shows the word frequency from the perspective of mandatory level in GRI and SASB Standards, also including the frequency of representative words at each level. For the level of requirements, recommendations and options, the representative words are respectively *shall*, *should* and *can*.

Besides the textual analysis, we conduct a case analysis of the original text of SASB and GRI standards, which is implemented into specific topics from three industries: Agriculture, Oil and Gas, and Coal, comprehensively covering all industry standards updated after the 2020 revision of the GRI standards. In addition to simply comparing the original standards, we

also select comparable companies in their respective industries and compare their ESG reports for 2021 with the disclosure indicators required. A detailed comparison table is available in Appendix, and the core content is extracted below. The results of the case study focus on the difference in the scope of indicator alternatives, while other relevant differences are about the classification and definition of indicators.

Table 4: Comparison between Specific Indicators of GRI and SASB Standards

Characteristics	Topic	ESG Disclosure Standards		
		GRI (2016)	GRI (2020)	SASB (2018)
Scope of Indicator Alternatives	GHG Emission	GRI305-4: CO ₂ , CH ₄ , N ₂ O, HFCs, PFCs, SF ₆ , NF ₃ , or All	GRI305-4: CO ₂ , CH ₄ , N ₂ O, HFCs, PFCs, SF ₆ , NF ₃ , or All	Oil Exploration: CO ₂ , CH ₄ , N ₂ O, HFCs, PFCs, SF ₆ , and NF ₃
	Air Emission	GRI305-7: NO _x , SO _x , POP, VOC, HAP, PM, other standards	GRI305-7: NO _x , SO _x , POP, VOC, HAP, PM, other standards	EM-EP-120a.1: NO _x , SO _x , VOCs, PM ₁₀
Classification of Indicators	Water	GRI303-3: Surface, Ground, Sea, Produced, Third-party	GRI303: Surface, Ground, Sea, Produced, Fresh, Others	EM-EP-140a.1: Fresh, Ground, Surface
	Waste	GRI306: Oil, Fuel, Wastes, Chemicals(soil and water), Others	N.A.	EM-EP-160a.2: Soil and Water Spills
Definition of Indicators	Waste	N.A.	N.A.	EM-EP-160a.2: Definition of Spills(>1bbl) and Scope (include and exclude)

Note: This table shows the difference in specifically required indicators of GRI and SASB Standards, covering topics of GHG emission, air emission, water and waste, which further concluded into three characteristics, the scope of indicator alternatives, classification and definition of indicators. The GRI Standards has two versions, 2016 and 2020, while the SASB Standards is the version 2018.

Among the two topics of GHG emission and air emission, the scope of indicator alternatives demonstrates a clear difference between the two standards. In terms of GHG emission, the SASB report specifies the types of GHG to be reported, and requires companies to disclose the seven gases of carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride (SF₆), and nitrogen trifluoride (NF₃) in strict accordance with the requirements of the Kyoto Protocol. However, the GRI Standards

does not mandate the scope of greenhouse gases, and companies have the autonomy to choose the types of gases to report, and only need to mention the reason behind them. In terms of air emission, the SASB has given requirements for NO_x, SO_x, VOCs and PM₁₀. In addition to the four SASB requirements, the GRI gives a broader range of significant gases, adding persistent organic pollutants (POPs) and hazardous air pollutants (HAPs), and pointing out that other standard categories of gas emissions identified in the relevant regulations can also be included in the usage.

In the topics of Water and Waste, GRI and SASB Standards reflect different classifications of disclosure. In the classification of water, SASB only requires the distinction between freshwater and non-freshwater, while GRI has a more diverse classification of water resources, including surface water, groundwater, seawater, produced water, freshwater, and others. In the classification of waste, SASB indicates that companies shall disclose spills only to soil and water separately. However, in addition to distinguishing between spills to water and spills to soil, GRI also distinguishes between oil spills, fuel spills, waste leaks, chemical leaks, and other leaks as defined by the organization. Companies can define their own classification of other water and other leaks and have greater discretion in determining the specific contents of the disclosure items.

The differences between GRI and SASB Standards in the definition of indicators are also reflected in the specific indicators in the Waste topic. SASB specifies requirements for the scope of releases in the disclosure of spills: “the scope of releases include above-ground and below-ground facilities, Sabotage, earthquakes, or other events outside operational control, Company-owned and operated transport, Leakage over time, which is counted once at the time it is identified, and Spills are defined as greater than 1bbl (42 U.S. gallons or 159 liters). Spills include those that reached the environment and exclude spills that were contained within impermeable secondary containment”. It can be seen that the SASB Standards provides a clear definition and boundary for spills in terms of quantity, while the GRI Standards does not have such a statement. This is not an isolated case. Therefore, it can be inferred that SASB is more explicit in its definition of some indicators, reflecting the attribute of being stringent.

Based on the word frequency analysis of the quantitative and mandatory levels, as well as the case studies of the standards text, the flexibility level of GRI and SASB Standards is confirmed, i.e. GRI is more flexible and SASB is more stringent. The SASB Standards is

more quantitative and mandatory, and defines the definition, connotation and subdivision of indicators, which deprives companies of the right to choose and helps them to make the decision of choosing specific indicators, and improves the efficiency of disclosure and comparability of indicators, which further facilitates investors to understand ESG information and make investment decisions. This further facilitates investors to understand ESG information and make investment decisions. This is also in line with the original purpose of its establishment, which is “to develop and disseminate sustainability accounting standards that help public corporations disclose material, decision-useful information to investors”.

From the analysis of the two attributes of materiality and flexibility, we believe that the GRI Standards with double-materiality and flexibility tends to reflect the notion of “equity”, while the SASB Standards with a focus on financial-materiality and having the property of strictness, reflects more of “efficiency”. The specific relationship between their materiality and flexibility attributes and equity and efficiency is clearly stated in the section on deductive reasoning. It must be acknowledged that in the field of ESG disclosure standards, and indeed in all other fields, it is difficult to generalize an absolute representation of equity and efficiency. Therefore, here we believe that both the GRI and SASB Standards reflect the game between the two value propositions. However, in the context of ESG disclosure standards, GRI clearly tilts to the equity side of the scale, while SASB slides to the other side.

2.3 Hypotheses

Based on the qualitative and quantitative analysis above and relevant case studies, this part is intended to draw 4 important hypotheses in regard to 4 channels in the ESG system respectively, which are specifically cost of disclosure, ESG rating divergence, cost of debt and carbon emission. These hypotheses are closely linked to the later empirical analysis.

2.3.1 Cost of Disclosure

As is shown in the study of [Machado et al. \(2021\)](#), the GRI Standards is popular-used for its attention to all-around materiality. If the information omitted, misstated or covered up will affect the investment decision of the capital provider, this information is considered to be important. If the information lacks financial substance, while having a profound impact on the sustainable development of the environment and society, this piece of information also cannot

be ignored by companies. This disclosure purpose reflects the concept of equity in GRI's comprehensive attention, but also virtually increases the cost of corporate disclosure. When a company references the GRI standards for ESG reporting, in-depth research and calculation about its operating activities in the environment, society and other aspects are essential. A considerable amount of time and money are the disclosure costs brought about by the GRI standard.

In contrast, SASB focuses on single materiality, so it only requires companies to disclose sustainable development information that will directly affect the investment decisions of capital providers, which will significantly reduce the disclosure costs of companies. Companies only need to report in accordance with the lower requirements of SASB, thereby saving costs for generating ESG reports. The low cost of disclosure also reflects the important feature of SASB's emphasis on "efficiency". From this, we draw the following hypothesis 1:

H1: Holding all other conditions constant, companies using the GRI standards will have higher disclosure costs, while companies using SASB will have lower disclosure costs.

2.3.2 Divergence of ESG rating

As is shown by [Cornell \(2021\)](#), ESG rating scores are likely to have a significant positive impact on the cost of capital through theories such as investor preferences. As the most important reference for rating agencies, companies' ESG disclosure will undoubtedly have a profound impact on their ESG ratings. Specifically, the GRI standards requires a comprehensive disclosure based on the value proposition of double materiality. So rating agencies can gain a larger picture of the company from multiple perspectives, thus facilitating the reduction of rating divergence between different agencies on the same company. Therefore, the cost of capital of companies will also be further reduced, which coincides with the findings of [Plumlee et al. \(2015\)](#).

Adversely, the SASB Standards attaches importance to the concept of efficiency, which will make the ESG reporting of the company only contain important information for the investment decision of the capital provider, contrary to the ESG rating agencies' value of paying attention to multiple interests. Therefore, when rating agencies refer to the ESG report that only applies to SASB, they may inevitably speculate about the company's impact on other

aspects due to the lack of key comparable information, thus exacerbating the divergence between different rating agencies on the same company. Hence, we draw the following hypothesis 2:

H2: Holding all other conditions constant, the divergence of ESG ratings obtained by companies using the GRI Standards will be smaller, and the divergence of ESG ratings obtained by companies using the SASB Standards will be larger.

2.3.3 Indirect Finance

Companies' ESG reports are one of the most important references when the bank decides the lending rate to the company. When making loan decisions, banks may pay less attention to the impact of companies on the environment, society and other aspects than ESG rating agencies. On the contrary, as the most typical representative of debt financiers, they care most about the financial and credit situation of companies, which is consistent with the principle that SASB attaches financial materiality. In the theory of information asymmetry, [Berger et al. \(2005\)](#) believe that banks are not willing to provide loans to companies with less transparent information, especially large banks. Therefore, we can deduce that the ESG report provided by companies using SASB Standards will be able to provide denser and more relevant information to banks, and assist them not only in the interpretation and comparison of the ESG data but also the efficiency in making loan decisions, so those companies may be benefited by a relatively low-interest rate.

In contrast, the ESG report provided by companies using the GRI Standards may consists of more redundant information from the perspective of bankers, which concern less about the impact on the environment and social equity. Benabou and Tirole, 2010, as well as Kitzmuelle and Shimshack, 2012 even found that managers who engage in ESG activities are suspected of spending company resources to enhance their personal reputation or take the opportunity to stabilize their own position. After the complex, boring and even inefficient information screening process, companies are likely to obtain relatively higher loan interest rates. From this, we draw the following hypothesis 3:

H3: Holding all other conditions constant, companies using the GRI Standards may obtain

higher bank loan interest rates, and companies using the SASB Standards may obtain lower loan interest rates.

2.3.4 Carbon emissions

As an ESG disclosure standard based on the mindset of double materiality, GRI was originally established to urge companies not only to disclose but also to manage and improve their ESG performance. The study by Cong et al. (2022) explores the relationship between ESG investments and carbon emissions in China and shows that a 1% increase in environmental investments would cause a 0.246% decrease in CO₂ emissions and a 0.558% decrease in carbon emission intensity. It can be deduced that companies adopting GRI for disclosure, in order to write a more comprehensive ESG report, undoubtedly need to implement the fundamental goal of protecting the ecological environment and taking social responsibility in their daily operations. In order to achieve this, one of the most important measures for companies is to reduce carbon emissions through methods including green production.

Contrarily, due to the nature of accounting standards, the SASB Standards based on the mindset of financial materiality propose fewer requirements for companies to disclose indicators related to pure eco-protection and human rights protection, and the motivation to take more social responsibility will also be weakened. Therefore, it is likely that there will be no significant difference between indicators such as carbon emissions before and after the adoption of SASB. From this, we draw the following hypothesis 4:

H4: Holding all other conditions constant, companies using the GRI Standards will significantly reduce their carbon emissions, while companies using SASB will not witness significant changes in carbon emissions.

3 Data

3.1 Firsthand Data from ESG reports

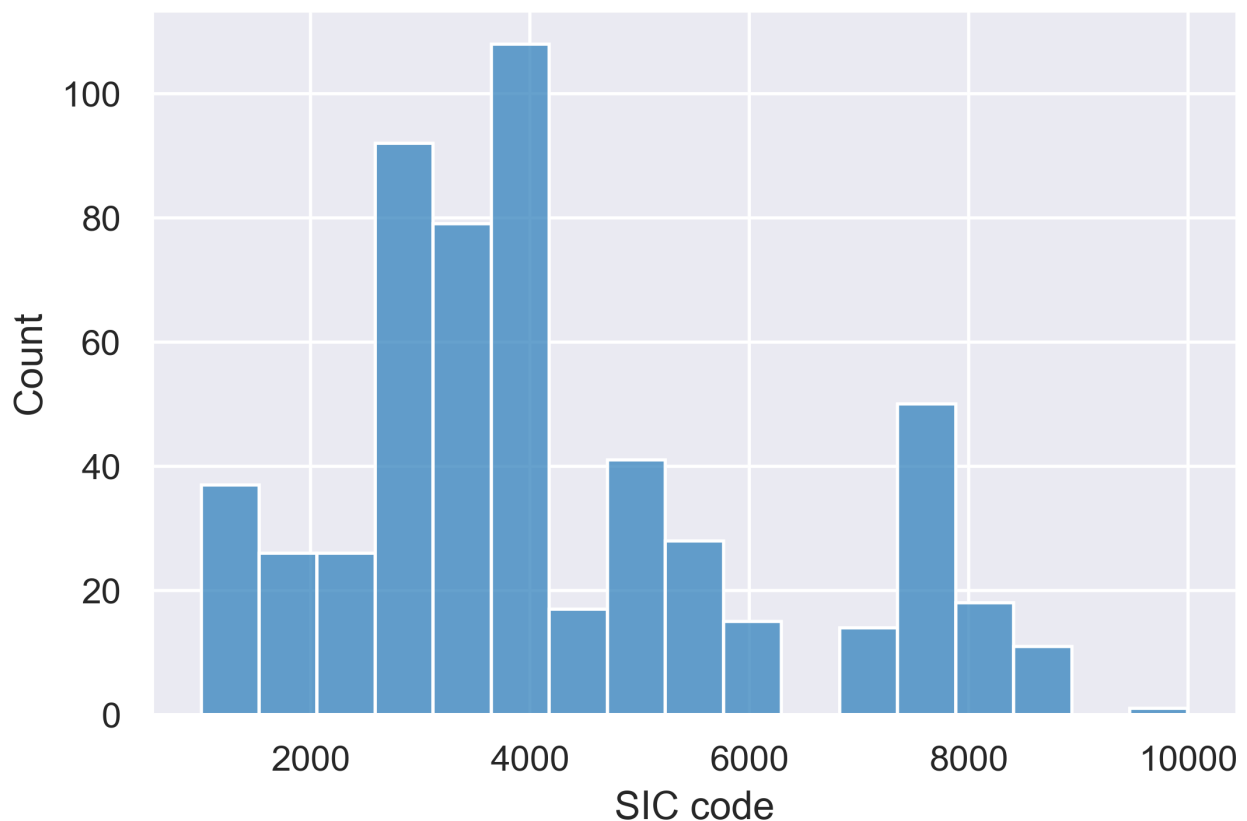
We use Python open source third-party libraries request and selenium to build a web crawler, obtain data sources through the Google search engine ^{*}, and crawl the ESG disclosure infor-

^{*}<https://www.responsibilityreports.com/>

mation of each company, including sustainable development report, corporate responsibility report, corporate social responsibility report and ESG report. A total of 1,000 companies were randomly selected for crawling. After filtering invalid data, the ESG disclosure reports of 597 sample companies were collected. Figure 6 shows the industry distribution of the sample companies, covering most industries in the standard industry classification, and the sample selection bias has little impact on the endogeneity brought on by the subsequent empirical study.

The basic attributes of the crawled report are extracted to obtain the reporting year, report page number, report file size and other fields. After text extraction in pdf reports using Python, mainstream standard keywords (including GRI, CDP, CDSB, SASB, TCFD, OECD, ISO26000, IIRC, VRF, and SDG) were matched, and the disclosure standards adopted in each company's reports were confirmed according to the frequency of standard keywords. After manual verification, the accuracy can reach 95%.

Figure 6: Industry Distribution of Firms



Note: This figure shows the distribution of ESG-report data from different sectors, which are denoted by the Standard Industrial Classification code from Compustat. The sample covers

the period from 2000-2021.

We use the python module “PyPDF2” to extract the ESG report PDF file size and pages of each company. After transforming PDF files to TXT files, we conduct text analysis to acquire information related to the text content. To retrieve which standard a specific company has adopted, we use the “in” method of string type in python, judging whether the standard names appear in the report files, such as “GRI”, “SASB”, “TCFD” etc., and count the number of standards adopted by each company. Furthermore, through a regular expression, which is the “re” module in python, we approximately retrieve how many effective statistics a company has disclosed in the report file. We set the rules of “defining effective statistics” and the accuracy rate can reach up to over 90%. The rules are as follows:

- (i) Exclude the numbers 1950 to 2050, since they can represent years.
- (ii) Exclude the invalid numbers including pages, serial numbers, and messy codes by setting whitespaces before and after the number we want.
- (iii) Include the valid numbers by adding those numbers with only one letter as a suffix, such as 100m, 66b, etc.
- (iv) Include the valid numbers containing the thousandth sign.

Out of a sample of 3,817 valid ESG reports that we obtained, as many as 2,889 reports chose the GRI standard for disclosure. Even if the retrieval threshold is raised, the number is as high as 2526, which shows the well-deserved dominant position of GRI in all major information disclosure standards, and the trend of convergence when enterprises choose disclosure standards. SASB has nearly 1,000 occurrences, ranking fourth among all criteria. But given its fairly short history, and the frequency with which CDP and SDG have been used together with GRI (which will be further discussed in the next part), this is certainly an impressive result.

The average number of disclosures per report using both criteria is not large and is indicative of the gradual convergence of standards. In addition, on average, each report file is about 9MB in size, contains about 55 pages, and contains about 242 data points. Using the median gives a better picture of the central tendency of these statistics, allowing for some outliers – typically, each report is about 5.8M in size, 46 pages long and contains 166 data points.

Table 5: Summary of ESG reports Statistics

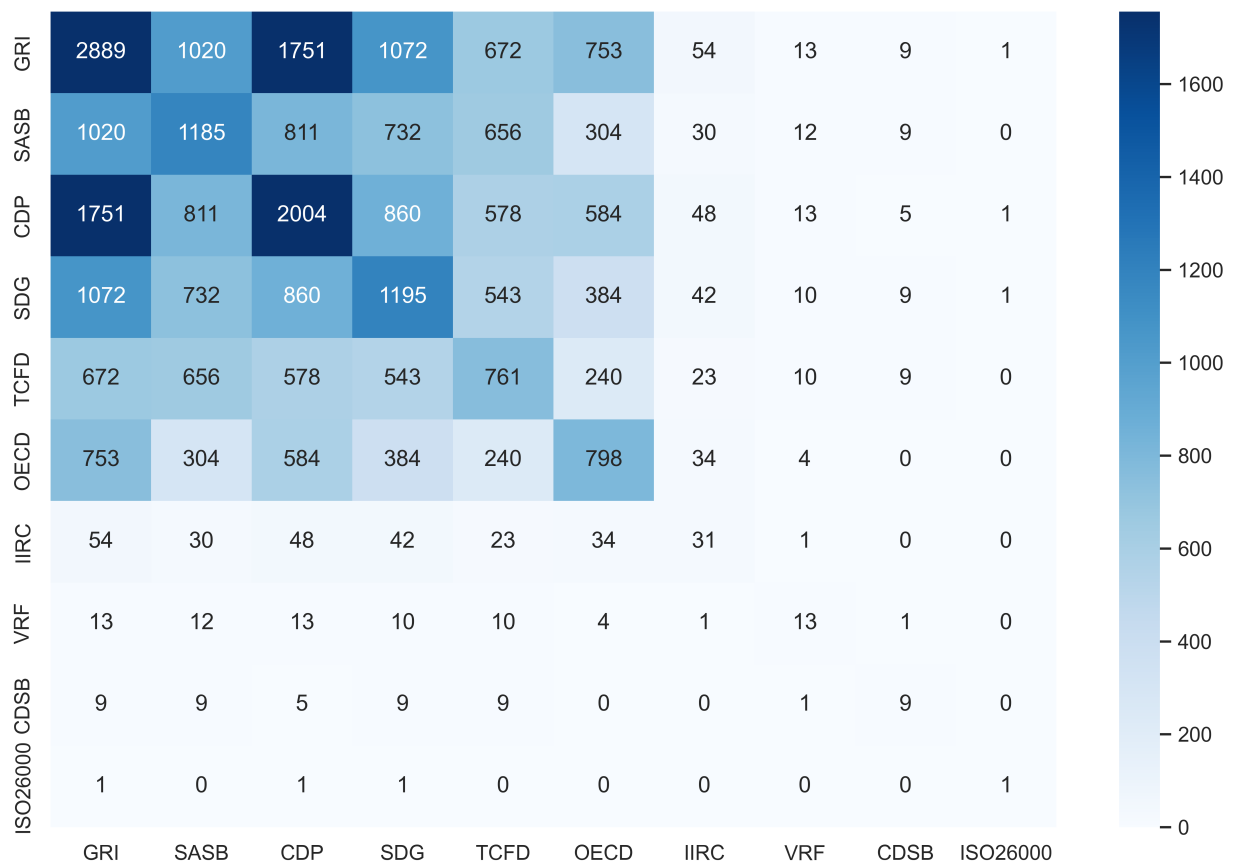
Panel A. Firms with ESG guideline response						
Guidelines	Reports with general guideline response			Reports with specific guideline response		
GRI	2,889			2,526		
SASB	1,185			976		
CDP	2,004			1,588		
SDG	1,195			1,049		
TCFD	761			607		
OECD	798			339		
IIRC	54			24		
VRF	13			7		
CDSB	9			4		
ISO26000	1			0		
OTHERS	519			0		
Total	3,817			3,070		
Panel B. Summary statistics						
Variable	N	Mean	SD	Min	p50	Max
FileSize(MB)	3,817	9.08	11.75	0.11	5.80	161.82
FilePages	3,809	55.47	40.83	1.00	46.00	376.00
GuideNums	3,809	2.05	1.61	0.00	2.00	7.00
Statistics	3,809	242.16	259.49	0.00	166.00	3,046.00

Note: This table reports summary statistics of data from ESG reports. Panel A shows the number of different ESG disclosure standards the firms use. Reports with general standards responses mean that firms only show which standard they use in one word when they make the framework and details of ESG reports. Reports with specific standard responses mean that the firms will make a table to show which pages correspond to which issues of the ESG disclosure standards specifically. The correspondence between the standard abbreviation and the full name is GRI-Global Reporting Initiative, SASB-Sustainability Accounting Standards Board, CDP-Carbon Disclosure Project, SDG-Sustainable Development Goals, TCFD-Task Force on Climate-related Financial Disclosures, OECD-Organization for Economic Cooperation and Development, IIRC-International Integrated Reporting Council, VRF-Value Reporting Foundation, CDSB-Climate Disclosure Standards Board, ISO26000-International Standard Organization 26000. OTHERS means that the firms don't show they use any standards mentioned above in their ESG reports. Panel B shows the summary statistics of four variables we con-

struct (numbers, averages, medians, minimums, maximums and standard deviations).

It is common for a report to adopt multiple standards at the same time, among which the combination of GRI with CDP and SDG is the most common. Given the common feature that these three standards focus on dual importance and are committed to sustainable development in the world, it is not surprising that companies choose both of them in their disclosure. In addition, SASB is used in conjunction with GRI more than 1000 times. Given the differences in the logic underlying the design of the two standards – GRI emphasizes equity and SASB emphasizes efficiency, this large number undoubtedly highlights their tendency to converge – it is best to ensure equity without losing efficiency.

Figure 7: Times of Standards Used Simultaneously



Note: Figure reports the number of intersections between two different ESG disclosure guidelines. An increase (decrease) in the depth of color signifies an increase (decrease) in the number of intersections.

3.2 ESG Ratings

ESG ratings first emerged in the 1980s as a way for investors to screen companies on environmental, social, and corporate governance performance. The earliest ESG rating agency, Eiris (merged with Vigeoin2015), was established in 1983 in France, and 7 years later, Kinder, Lydenberg, and Domini (KLD) was established in the USA. While initially catering to a highly specialized investor clientele, including faith-based organizations, the market for ESG ratings has widened dramatically, especially in the past decade. Because ESG ratings are an essential basis for most kinds of sustainable investing, the market for ESG ratings grew in parallel to sustainable investing. As sustainable investing transitioned from niche to mainstream, many early ESG rating providers were acquired by established financial data providers. For example, MSCI bought KLD in 2010, Morningstar acquired 40% of Sustainalytics in 2017, Moody’s bought Vigeo-Eiris in 2019, and S&P Global bought RobecoSAM in 2019.

We use data from four different ESG rating providers: MSCI KLD, Sustainalytics, Refinitiv, and S&P Global. We include KLD because it is the data set that has been used most frequently in academic studies. Our selection of the other raters is guided by market relevance. All the providers in our sample are widely recognized and used by sustainable finance professionals. We request access to the ratings, the underlying indicators, and documentation about the aggregation rules and measurement protocols of the indicators. The KLD data set does not provide an aggregate rating; it only provides binary indicators of “strengths” and “weaknesses.” We create an aggregate rating for KLD by following the procedure that is chosen in most academic studies—namely, summing all strengths and subtracting all weaknesses (Lins et al., 2017).

Table 6 provides descriptive statistics of the aggregate ratings and their sample characteristics. The sample covers the period from 1999 to 2021. Because most of the academic literature to date relies on KLD data, it is important to include it in our study. Panel A shows the full sample, where the number of firms ranges from 413 to 575. Panel B shows the common sample of 315 firms. The mean and median ESG ratings are higher in the common sample for all providers, indicating that the balanced sample tends to drop lower-performing companies. For our further analysis, we normalize the common sample to have zero mean and unit variance in the cross-section.

Table 6: Summary Statistics of ESG ratings

Panel A. Full sample							
	Firms	Firm_year	Mean	SD	Min	Median	Max
Sustainalytics	415	1,918	59.94	8.57	41.00	60.00	82.74
S&P Global	560	2,582	33.54	21.08	3.34	28.09	90.75
KLD	413	1,901	2.34	1.42	1.86	7.32	11.28
Refinitiv	575	3,419	0.78	0.22	0.06	0.88	0.99
Panel B. Common sample							
	Firms	Firm_year	Mean	SD	Min	Median	Max
Sustainalytics	315	1,076	60.37	8.34	41.36	60.63	80.27
S&P Global	315	1,076	38.04	20.61	5.42	32.75	89.83
KLD	315	1,076	2.65	1.24	2.14	7.74	11.23
Refinitiv	315	1,076	0.84	0.17	0.06	0.91	0.99

Note: This table shows descriptive statistics of the aggregate ratings (ESG level) for the four rating agencies. Panel A shows the data for the full sample and Panel B for the common sample.

3.3 Carbon Emissions

The data on the firm's climate performance is source from S&P Trucost, which provides annual information on firm-level carbon emissions in tons of carbon dioxide (CO₂) equivalent (tCO₂e). The Trucost data is matched to the stock-level information by cusip, ISIN and company names. Scope 1 GHG emissions cover direct emissions from owned or controlled sources by the firm. Scope 2 GHG emissions cover indirect emissions from the generation of purchased electricity, steam, heating and cooling consumed by the reporting company. Scope 3 GHG emissions include all other indirect emissions that occur in a company's value chain. Scope 1 emissions are fully controlled by the firm, while scope 3 emissions are less in the firm's control. Scope 3 emissions can be further divided into upstream and downstream emissions based on the supply chain. Because the scope 3 total downstream emission data start in 2017, we only study the upstream for scope 3 emissions. The key measure of the carbon-transition risk is carbon intensities, that is carbon emissions scaled by the fiscal year-end sales for the fiscal year. We consider a few alternative variables, including log carbon emissions and log year-on-year change of carbon emissions proposed in [Bolton & Kacperczyk \(2021\)](#). If the latest carbon data for the latest fiscal year is not released yet, we fill in the missing variables, emissions, growth, or carbon intensities, with the latest available number.

3.4 Bank Loan

We obtain corporate loan data from LPC DealScan over the 2000 to 2020 period. This database provides information on loan characteristics such as yield spreads, maturity, facility amount, purpose, type, and other contract terms. Most loans covered in DealScan are syndicated. Following the literature, we focus our attention on the lead arranger(s) rather than syndicate participants, because the lead bank(s) assume an active role in originating the loan contract.

3.5 Financial Statements

Firms' stock and financial data are available from CRSP and Compustat, respectively. The [Chava & Roberts \(2008\)](#) link file enables us to combine the sample of loans from DealScan and borrowers' financial information from CRSP and Compustat into a single dataset. We restrict our sample to U.S. stocks.

4 Empirical Analysis

4.1 Cost of Disclosure

In this part, we explore the impact of ESG disclosure standards on firms' disclosure behavior. A key assumption is that the more data a standard requires to be disclosed, the greater the cost of the disclosure will be. Most institutions only conduct research on ESG disclosure standards through text analysis of standard documents, but GRI and SASB have different frameworks, which makes their conclusion to be not completely credible. Therefore, besides the previous qualitative and quantitative analysis, we choose data using ESG disclosure standards at the company level. By analyzing the characteristics of different ESG disclosure standards and the ESG report itself, we reveal how standards affect corporate ESG reporting. We first conduct a T-test for differences between groups, and the results are shown in Table 7. Then we make Logit and Probit regression to explore the reasons why a firm chooses certain ESG disclosure standards.

Table 7: Characteristics of ESG Reports and ESG Disclosure Standards

	N	GRI	GRI (Controls)	N	SASB	SASB (Controls)	diff. in means
		(1)	(2)		(3)	(4)	(5)
Filesize	1368	7.69		215	10.72		-3.03*** (-4.46)
	767		6.51	82		10.10	-3.59*** (-3.32)
Filepages	1368	62.39		215	43.12		19.27*** (6.54)
	767		53.08	82		35.15	17.94*** (4.06)
Guidenums	1368	1.91		215	2.35		-0.44*** (-6.42)
	767		1.13	82		1.39	-0.26*** (-6.09)
Statistics	1368	304.26		215	132.04		172.23*** (8.56)
	767		248.05	82		105.50	142.55*** (4.84)

Note: Column (1) provides summary statistics for GRI users in the year releasing ESG reports. In Column (2), we control other guidelines' effects and only choose the reports that only use GRI for data disclosure. Columns (3) and (4) follow the rule and we show the T-test in Column (5).

Regardless of whether the uniqueness of the standard selected in the ESG report is controlled or not, the reports using GRI for disclosure show significantly smaller file sizes, more report pages, and more data points. This is intuitive, indicating that the GRI standard with dual importance requires more detailed disclosure requirements, and companies need to include more aspects in their reports. In contrast, enterprises that choose SASB only focus on the financial and economic value and tend to add more pictures to enrich the content of the report, which is undoubtedly a cheaper and more efficient behavior for enterprises. This significant distinction directly reflects the difference in disclosure costs between the two standards, which will be further discussed in the next section.

As mentioned above, the SASB, which focuses only on single materiality, leads to lower disclosure costs for companies. This implies the case that in the past year, a company might fail to make enough efforts in ESG, so it will be more motivated to choose the low-cost and efficient SASB standard for disclosure, in order to obtain a financial and economic improvement

in a short time. In contrast, enterprises in the mature stage are more inclined to use GRI, which represents double materiality and equity. The above results also perfectly confirm the validity of hypothesis 1. However, can these two types of companies achieve their aspirations? This will be the focus of the next section.

$$\begin{aligned}
Choice_{i,t} = & \alpha_0 + \alpha_1 LESGmean + \alpha_2 LESGN + \alpha_3 Size \\
& + \alpha_4 BM + \alpha_5 Leverage + \alpha_6 TobinQ \\
& + \alpha_7 ROA + \alpha_8 MNC + FixedEffects + \tau_{i,t}
\end{aligned} \tag{1}$$

Table 8: Regressions of Firms' Standards Choices on ESG and Corresponding Financial Features

VARIABLES	(1) Logit Choice (SASB = 1; GRI = 0)	(2) Probit Choice (SASB = 1; GRI = 0)
<i>LESGmean</i>	-0.030*** (-4.51)	-0.016*** (-5.07)
<i>LESGN</i>	-0.510*** (-3.49)	-0.281*** (-4.06)
<i>Size</i>	0.012 (0.17)	0.007 (0.20)
<i>BM</i>	3.051*** (2.92)	1.770*** (3.13)
<i>Leverage</i>	3.794*** (5.66)	1.981*** (5.74)
<i>TobinQ</i>	0.224*** (4.53)	0.128*** (4.51)
<i>ROA</i>	-0.038*** (-3.71)	-0.021*** (-3.81)
<i>MNC</i>	0.456* (1.91)	0.210* (1.79)
Constant	-4.425*** (-3.52)	-2.514*** (-3.81)
Year FE	Yes	Yes
Industry FE	Yes	Yes
Observations	1,668	1,668
Pseudo R2	0.144	0.15

Note: The table reports the regression results of firms' standard choices on ESG and financial features. The dependent variable takes one when the firm chooses SASB for ESG disclosure, while zero when GRI is chosen. Column (1) documents the estimates based on the Logit model and Column (2) is the Probit model. Industry (Fama-French 49 industries), time fixed effects are controlled in the model. The standard errors are clustered by firm and year. T-statistics are reported in parentheses. The symbols *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

4.2 ESG Rating Divergence

This part will mainly explore the impact of disclosure standards on corporate direct finance. Many studies have confirmed the negative impact of ESG rating divergence on investors' investment decisions in the stock market (Berg et al., 2022). Based on previous findings, we will further explore the moderating effect of disclosure standards on the divergence of ESG ratings, which may reveal its impact on corporate direct finance.

Drawing on studies on investor disagreement in financial markets, two main reasons for divergence are differences in information sets and differences in interpretations of the information (Cookson & Niessner, 2020). Financial disclosures usually reduce dispersion among equity analysts, as well as credit rating analysts owing to a widespread agreement on the meaning of financial information (Morgan, 2002; Bonsall & Miller, 2017; Akins, 2018), while ESG disclosures are likely to exacerbate dispersion among ESG rating analysts because of little agreement on how to interpret and judge the meaning of ESG performance disclosures. According to this point of view, Christensen et al. (2022) find empirical evidence that greater ESG disclosure gives rise to greater ESG rating disagreement. Nevertheless, what about quantitative ESG disclosure?

To test the hypothesis that a greater quantitative measure of ESG disclosure will give rise to greater divergence of ESG ratings, the main identification model is as follows:

$$ESGDivergence_{i,t} = \gamma_0 + \gamma_1 ESGStats_{i,t} + \sum \gamma_k Control_{i,t} + FixedEffects + \tau_{i,t} \quad (2)$$

where i indexes firm, t indexes year. $ESGDivergence_{i,t}$ is the standard deviation of ESG ratings of firm i in year t , and $\gamma_1 ESGStats_{i,t}$ is the quantified indicators about ESG disclosure in firms' ESG reports. We predict that the coefficient of interest γ_1 will be significantly positive. $Control_{i,t}$ contains a vector of firm-level control variables, including the average ESG rating of a firm received from different rating agencies ($ESGmean$), the number of rating agencies following the firm ($ESGN$), total assets ($Size$), capital structure ($Leverage$), profitability (ROA), book-to-market ratio (BM), Tobin's Q ($TobinQ$), and an indicator variable for multinational corporations (MNC). Fixed effects consist of firm, and year. For the robustness of the results, we use industry, year fixed effects. Standard errors are clustered at the firm level.

Table 9: Effect of ESG Quantitative Disclosure Level on ESG Rating Divergence

VARIABLES	(1) ESGDivergence	(2) ESGDivergence	(3) ESGDivergence
<i>ESGStats</i>	1.171*** (5.07)	0.808*** (3.95)	0.835*** (4.18)
<i>ESGmean</i>	-0.028 (-1.44)	-0.004 (-0.22)	0.007 (0.40)
<i>ESGnum</i>	0.530*** (10.51)	1.076*** (10.69)	1.081*** (11.29)
<i>Size</i>	0.313 (1.58)	-0.461** (-2.32)	-0.577*** (-3.06)
<i>BM</i>	-6.942** (-2.05)	1.194 (0.38)	0.605 (0.20)
<i>Leverage</i>	-5.313*** (-3.01)	-0.246 (-0.15)	-0.047 (-0.03)
<i>TobinQ</i>	-1.150*** (-4.66)	0.146 (0.64)	0.003 (0.01)
<i>ROA</i>	0.179*** (5.43)	0.081*** (2.68)	0.069** (2.32)
<i>MNC</i>	1.893*** (3.38)	2.040*** (3.86)	2.055*** (4.08)
Constant	13.254*** (3.25)	-9.071** (-2.34)	-7.702** (-2.05)
Year FE	No	Yes	Yes
Firm FE	No	Yes	No
Industry FE	No	No	Yes
Clustering	Yes	Yes	Yes
Observations	1,760	1,759	1,760
Adjusted R-squared	0.111	0.363	0.348

Note: The table reports the regression results of the effect of ESG quantitative disclosure on ESG rating divergence. The dependent variable is the ESG rating divergence of four ESG rating agencies, which is measured by standard deviation. Columns (1) to (3) report three specifications of different fixed effects. The standard errors are clustered by firm and year. T-statistics are reported in parentheses. The symbols *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

No matter how the fixed effects model including year, company, or industry is designed, it

can be observed that the number of data points in the ESG report has a significant positive effect on the rating divergence. The original intention of various ESG disclosure standards, especially GRI, is to make the ESG data of enterprises transparent, standardized, and scientific, to reduce the difference in the rating scores of the same enterprise by different rating agencies and reduce the obstacles in investors' decision-making. Therefore, it can be assumed that the use of GRI and other disclosure standards by companies will have a significant moderating effect on their ESG rating divergence.

$$\begin{aligned}
ESGDivergence_{i,t} = & \gamma_0 + \gamma_1 ESGStats_{i,t} \times ESGGuides_{i,t} \\
& + \gamma_2 ESGStats_{i,t} \\
& + \gamma_3 ESGGuides_{i,t} + \sum \gamma_k Control_{i,t} \\
& + FixedEffects + \tau_{i,t}
\end{aligned} \tag{3}$$

where $ESGGuides_{i,t}$ is a dummy variable equal to 1 for firms that use ESG guidance (only GRI, only SASB or both) and 0 otherwise. $ESGDivergence_{i,t}$, $ESGStats_{i,t}$, $Control_{i,t}$, and $FixedEffects$ are the same as in Equation (3). γ_1 is the coefficient of interest, and if it is predicted to be significantly negative, it means that the introduction of ESG guidelines weakens the effect quantitative measure on ESG divergence.

Table 10: Effect of ESG Quantitative Disclosure Level on ESG Rating Divergence Using the Regulation of Disclosure Standards

VARIABLES	(1) ESGDivergence	(2) ESGDivergence	(3) ESGDivergence	(4) ESGDivergence
<i>ESGStats*ESGGuide(GRI)</i>	-1.406*** (-2.60)	-1.706*** (-3.21)		
<i>ESGStats*ESGGuide(SASB)</i>			0.781 (1.29)	1.283** (2.13)
<i>ESGStats</i>	1.384*** (4.92)	1.404*** (5.06)	1.115*** (4.48)	1.129*** (4.60)
<i>ESGGuide</i>	0.203 (0.40)	0.182 (0.36)	1.273* (1.77)	0.891 (1.26)
<i>ESGmean</i>	-0.113*** (-5.40)	-0.097*** (-4.84)	-0.113*** (-5.40)	-0.095*** (-4.73)
<i>ESGnum</i>	1.071*** (10.55)	1.073*** (11.05)	1.072*** (10.59)	1.076*** (11.16)
<i>Size</i>	1.002*** (4.84)	0.957*** (4.92)	1.053*** (5.10)	0.994*** (5.11)
<i>BM</i>	1.946 (0.56)	2.270 (0.67)	1.466 (0.42)	1.892 (0.55)
<i>Leverage</i>	4.281** (2.33)	5.399*** (3.05)	3.965** (2.15)	5.284*** (2.98)
<i>TobinQ</i>	0.270 (1.05)	0.118 (0.46)	0.233 (0.91)	0.094 (0.37)
<i>ROA</i>	0.109*** (3.25)	0.102*** (3.04)	0.112*** (3.33)	0.104*** (3.11)
<i>MNC</i>	2.716*** (4.61)	2.451*** (4.35)	2.727*** (4.62)	2.449*** (4.34)
Constant	16.040*** (3.86)	15.885*** (3.91)	15.891*** (3.85)	15.638*** (3.87)
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	No	Yes	No
Industry FE	No	Yes	No	Yes
Clustering	Yes	Yes	Yes	Yes
Observations	1,759	1,760	1,759	1,760
Adjusted R-squared	0.305	0.282	0.304	0.28

Note: The table reports the regression results of the effect of ESG quantitative disclosure on ESG rating divergence, using the regulation of disclosure guidance. The dependent variable is the ESG rating divergence of four ESG rating agencies, which is measured by standard deviation. The main independent variable in Columns (1) and (2) is the interaction between ESG statistics and GRI uses, while Columns (3) and (4) show SASB's. All columns report two different specifications of different fixed effects. Industry (Fama-French 49 industries) or firm fixed effect and time fixed effects are controlled in the model. The standard errors are clustered by firm and year. T-statistics are reported in parentheses. The symbols *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

It can be observed that by adopting the GRI Standards, the divergence of ESG rating will

indeed significantly reduce., while the use of the SASB Standards even tends to exacerbate the divergence. Moreover, GRI may enhance the differentiation of ESG ratings between separate companies, which is its another advantage over SASB and makes direct investors less confused about the rating results. The above analysis drives to an interesting conclusion that the efficiency-focused SASB does lack efficiency in the field of direct finance, outperformed by the equity-focused GRI completely.

Table 11: Horizontal Comparison of the ESG Rating Divergence

	GRI	SASB	GRI & SASB
Global variance	0.532	0.03	0.468
Single variance	0.082	0.153	0.118

Note: The Global variance indicates the level of ESG rating divergence among different companies; the Single variance indicates the level of ESG rating divergence from different rating agencies scoring the same company. It can be concluded that the GRI Standards outperforms SASB in both cases.

The performance of GRI and SASB at the rating level is compared based on the ESG disclosure standards used by the company and the ESG scores of the four rating agencies. Horizontally, the average rating score of each company is taken to calculate the distribution variance of the average score of all companies. The results show that the average rating span and variance of the companies using the GRI Standards are larger, indicating that the company ESG level differentiation brought by the GRI Standards is higher. Vertically, the variance of the rating scores of different rating agencies of each company is calculated and then the average is taken. The results indicate that the score variance of different rating agencies for companies using the GRI Standards is smaller, more similar, and less confusing for investors. This not only confirms the validity of hypothesis 2, but further forms an original and important conclusion of our paper.

4.3 Indirect Finance

We examine whether a firm with high ESG credibility is more likely to receive a loan from a bank. For this test, we first identify a sample of firms that receive a bank loan (i.e., subject firms) during the sample period. For each subject firm, we create a control group by choosing

firms that operate in the same industry and have comparable firm size, leverage, and market-to-book ratio (This matching process is similar to the one used in [Bernstein et al. \(2019\)](#)). These processes result in 275 subject firms and 314 control firms in our sample. An assumption behind this methodology is that both subject and control groups have similar capital policies, financing capabilities, and preferences for bank credit.

To better establish a potential causal relationship between borrower ESG practices and bank lending, we perform panel analyses using the introduction of ESG disclosure standards as shocks to how banks evaluate firm risks related to ESG. For a given firm-rebalance pair, we first require the firm included in the ESG guidance at least two years both before and after the rebalance event. Then, we only choose loan origination data closer to the event to better estimate the pure effects of the use of ESG guidance. The empirical specifications are as follows:

$$\begin{aligned} LnSpread_{i,j,t} = & \alpha + \beta LnESGlast_{i,t} + \varsigma X_{i,t} + \tau Y_t + \gamma Z_{i,j,t} \\ & + I_{ind} + \nu_i + \chi_j + \xi_{i,j,t} \end{aligned} \quad (4)$$

In the model, the dependent variable is the natural logarithm of the loan spread over the London Interbank Offered Rate (LIBOR) charged by a bank. $LnESGlast_{i,t}$ (borrower's ESG scores of the lag period) is the key independent variable. X is the vector of borrowers' traits, which contains the log of total assets, market-to-book, cash, book leverage, and credit rating. Y and Z respectively denote macroeconomic variables (i.e., the natural logarithm of M2) and the vector of loan traits that include relationship length, the maximum of loan amount, collateral. We also control for the industry(I), borrower(ν), and lender(χ) fixed effects, and the standard errors are clustered by borrower and year.

Table 12: Regressions of Loan Spreads on Firms' ESG Ratings

	(1) All Ln(Spread)	(2) GRI Ln(Spread)	(3) SASB Ln(Spread)
Ln(ESG)	-1.192*** (-3.66)	-0.878* (-2.23)	-1.307*** (-3.51)
Collateral	-0.094 (-0.77)	-0.072 (-0.91)	-0.122 (-1.07)
Relationship Length	-0.000*** (-4.30)	-0.001*** (-6.76)	-0.000*** (-3.40)
Loan_max	-0.000*** (-3.89)	-0.000** (-3.08)	-0.001 (-0.97)
Size	0.267 (0.79)	0.249 (1.20)	0.219 (0.69)
LnCash	-0.055 (-0.55)	0.033 (0.52)	-0.064 (-0.64)
BM	0.325* (1.92)	0.584*** (4.44)	0.329* (1.89)
Leverage	0.273** (2.32)	0.743** (2.27)	0.366** (2.43)
Credit Rating	-0.012** (-2.17)	-0.014* (-2.06)	-0.012** (-2.16)
LnM2	-0.552* (-2.04)	-0.681*** (-5.00)	-0.502* (-1.93)
Clustering	Yes	Yes	Yes
Loan Characteristics	Yes	Yes	Yes
Industry FEs	Yes	Yes	Yes
Borrower FEs	Yes	Yes	Yes
Lender FEs	Yes	Yes	Yes
Observations	1,177	976	201
Adjusted R-squared	0.475	0.392	0.501

Note: The table reports the regression results of loan spreads on firms' ESG ratings. The dependent variable is loan spreads. Column (1) reports the estimates based on the whole sample of ESG disclosure guideline use. Column (2) reports the estimates based on the subsample of GRI users, while Column (3) reports SASB users. Industry (Fama-French 49 industries), borrower, lender and time-fixed effects are controlled in the model. The standard errors are clustered by firm and year. T-statistics are reported in parentheses. The symbols *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

The results in this part strongly prove that the SASB Standards alleviates the indirect

financing constraints of enterprises – companies using the SASB Standards enjoy significantly lower bank loan interest rates, which is indeed in line with the original intention of the SASB Standards. Through the most efficient ESG information disclosure, the companies in a short period of time obtain economic and financial improvement. Therefore, small and medium-sized companies in the growth stage can indeed adopt this quite efficient standard for information disclosure in the short term, and their previous decisions are scientific and reasonable. The above results also perfectly confirm the validity of hypothesis 3.

4.4 Carbon Emissions

In this part, we explore the impact of ESG disclosure standards on corporate carbon emissions. Since GRI and SASB have different disclosure frameworks and core indicators, they require different data disclosures. In other words, the SASB Standards may require more disclosure related to finance, while the GRI requires more data related to ESG itself, with carbon emissions as one of the most important ones. We hypothesize that there is an oversight mechanism when companies are required to disclose more environmental data. When a company needs to disclose data to the public, they intentionally send good signals. So companies that are more transparent tend to do better financially and environmentally. But we believe that the way data is disclosed can affect the effectiveness of this mechanism. For example, because GRI needs to disclose more environmental data, GRI will play a better role than SASB in controlling carbon emissions. Therefore, we design the following model:

$$\begin{aligned} Carbon_{it} = & \alpha_0 + \beta_1 post \times treatment + \beta_2 post + \beta_3 treatment \\ & + \sum \beta_k Control_{i,t} + FixedEffects + \tau_{i,t} \end{aligned} \quad (5)$$

We perform difference-in-differences (DID) analyses. The *post* takes the value of one for the introduction of SASB or GRI in firms' ESG reports, and zero for those before the event. *Treatment* refers to whether firms use ESG disclosure standards or not. If it does, *treatment* takes the value of one, and zero when we can't find any standards in ESG reports. The coefficient of the interaction term captures what we care about.

Table 13 represents the main results. The coefficient on DID shows the baseline effect. The negative and significant coefficient estimate for the interaction term shows that the introduc-

tion of GRI has an effect on carbon emission reduction, while this effect is less significant or almost none in SASB. We only find evidence of total carbon emissions, instead of the growth rate of carbon emission of carbon intensity. This result is consistent with our hypothesis 4.

Table 13: Regressions of Carbon Emissions on the Introduction of ESG Disclosure Standards

Panel A: GRI									
Dependent variable	SCOPE 1			SCOPE 2			SCOPE 3		
	TOT	GROWTH	INT	TOT	GROWTH	INT	TOT	GROWTH	INT
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
GRI*Disclosure	-0.244** (-2.04)	0.054 (1.29)	-0.132 (-0.76)	-0.424*** (-4.75)	0.020 (0.33)	-0.559* (-1.84)	-0.195*** (-3.01)	0.113 (0.50)	-0.537 (-0.61)
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,043	1,628	2,043	2,019	1,607	2,019	2,595	2,111	2,595
Adjusted R-squared	0.339	0.188	0.047	0.496	0.174	0.072	0.417	0.045	0.295
Panel B: SASB									
Dependent variable	SCOPE 1			SCOPE 2			SCOPE 3		
	TOT	GROWTH	INT	TOT	GROWTH	INT	TOT	GROWTH	INT
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
SASB*Disclosure	-0.382 (-1.52)	0.025 (1.03)	-0.293 (-0.51)	-0.492* (-1.86)	0.055 (0.28)	-0.694 (-1.57)	-0.392 (-1.28)	0.323 (0.31)	-0.451 (-0.52)
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,043	1,628	2,043	2,019	2,019	1,607	2,595	2,111	2,595
Adjusted R-squared	0.074	0.098	0.042	0.140	0.024	0.071	0.087	0.025	0.262

Note: The table reports the regression results of carbon emissions on the introduction of different ESG disclosure standards. Panel A shows the GRI part, while Panel B shows the SASB. We divide dependent variables into three groups for scope 1, scope 2 and scope 3. The first column in each part reports the total emissions, the second report the growth rate of carbon emissions and the third one is about carbon emission intensity, which is emissions scaled by sales of firms. Firm and time-fixed effects are controlled in the model. The standard errors are clustered by firm and year. T-statistics are reported in parentheses. The symbols *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

5 Conclusion and Suggestions

This section will be divided into four parts. Firstly, we will provide a final summary of the conclusions drawn from the theoretical analysis and empirical results. The second part is our further exploration of the implications from the perspective of alignment. Material suggestions for internal designers and external users of the standard are provided in the third part. Finally, we will discuss the deficiencies of this study and the prospects of improvement.

5.1 Summary of Conclusions

From the study above, we can indeed draw significant differences between the two standards, GRI and SASB. While the former is always showing the concept of “equity” in its design and application process, the latter can be considered as the most typical representative of “efficiency”. Specifically, the GRI is famous for its double materiality, reflecting equity across species and even generations, but is less efficient due to its overly comprehensive disclosure requirements; the SASB, on the other hand, focuses only on the economic and financial materiality of the company, which is efficient for both the company as a discloser and external investors including banks, but lacks a focus on ESG factors such as the environment and society. The GRI’s flexibility in disclosure requirements, allowing companies to choose the items that best suit their circumstances, is indeed in line with the concept of equity; the SASB’s more stringent requirements mean that comparability is enhanced, providing external investors with more dense and effective information while also achieving the “efficiency first” principle.

In terms of disclosure costs, a company’s ESG report is likely to have more pages and more data points if it references the GRI, which entails higher disclosure costs; while a report using the SASB Standards, which has lower disclosure costs, is likely to be “fleshed out” with images. The difference in cost of disclosure between the two standards is further reflected in the characteristics of the companies that choose them. The GRI, with its higher disclosure costs but a greater demonstration of equity, is often preferred by large, mature companies; while the SASB, with its lower disclosure costs but greater emphasis on efficiency, is chosen by smaller, growing companies. In terms of ESG rating divergence, the GRI, by fitting the pluralistic values of ESG rating agencies, results in less ESG rating divergence for companies

using its criteria, and also increases the divergence of ESG ratings between companies, thereby facilitating direct finance in both the vertical and horizontal dimensions. This is an important manifestation of the “efficiency” of the equity-focused GRI, while also proving the lack of efficiency in the field of direct finance for the efficiency-focused SASB. This is one of our key findings, as almost no previous conclusion has been drawn from this perspective.

In terms of indirect finance, our empirical results provide strong evidence that SASB reduces the financing constraints of companies, which is consistent with its efficiency focus and reflects the wisdom of the decisions made by small companies to choose SASB to weather the difficulties in their growth phase. In contrast, the GRI does not have a significant effect in reducing financing constraints. In terms of carbon emissions, the GRI does fulfill its original promise of global cross-species and cross-generation equity and has been successful in promoting green production levels in companies that use it as a disclosure standard, while the SASB, which holds financial-materiality and efficiency-focused, has less significant results in this regard.

5.2 Further Refinement

Although the empirical results are not yet significant in a few areas (e.g. ESG ratings for direct finance), the above summary of findings has demonstrated the typical characteristics of GRI and SASB in terms of their respective equity and efficiency, and such a comparative analysis begs an important question: why exactly are we making such a comparison? To answer this question, we offer here a unified and rigorous response: the core purpose of comparing the two standards is to facilitate their alignment. By alignment, we do not simply add the two sets of standards together, but rather shape a more mature set of standards based on them, balancing equity and efficiency in the context of maximizing the combined internal and external value of the company. In fact, all ESG disclosure standards align with each other, but the reason for comparing only GRI and SASB is that they are each the most typical representatives of the two value propositions. Other standards are more or less biased towards one of these two and will be absorbed by the new comprehensive standard as these two standards align with each other.

Why is the alignment of standards, or the integration of equity and efficiency in the same standard inevitable? Firstly, from the perspective of the standards themselves, their funda-

mental purpose is to promote a higher level of sustainable development worldwide by urging companies to publish and manage ESG information in a scientific manner. If equity is lost, it will not reflect the importance attached to diverse values such as ecology, environment and social responsibility around the world; if efficiency is lost, companies will not have sufficient incentive to adopt the disclosure standard, and the goal of sustainable development will not be achieved in reality.

Secondly, from a company's perspective, it is also essential to have a set of disclosure standards that balance efficiency and equity to guide its operations. The example of the Industrial Revolution has demonstrated the negative environmental and social effects of the mindless pursuit of technological progress and profit enhancement. Each individual capitalist will not hesitate to choose profit maximization as his or her first priority in starting a company, which is certainly an expression of the pursuit of efficiency, but when all capitalists around the world make this choice, equity across species and generations will be completely broken. This is further evidence of the relationship between equity as primary and efficiency as secondary: equity without efficiency may still exist in nowhere, but efficiency without equity is ultimately transient. Equity with efficiency is the core goal we pursue, while efficiency with equity is always suspected of being utilitarian.

In the end, what value will such an alignment produce? When a more mature standard takes efficiency into account, companies will have lower disclosure costs when adopting it, direct and indirect financing will also be less constrained, and the economic value of the company will increase directly; when a more mature standard takes equity into account, companies will have significantly lower carbon emissions, for example, and the level of sustainability will increase worldwide. Overall, the internal and external value of the company is improved in an integrated way.

5.3 Material Suggestions

Having discussed the intrinsic meaning and need for alignment of standards, this section will look at possible initiatives to promote the maximization of internal and external corporate value in the context of the establishment of a market-based ESG system. The most important aspect of maximizing internal and external value for companies is to integrate ESG performance into the business design, which can of course rely on regulatory oversight and

management, but more importantly the establishment of market-based ESG systems. In a market-based ESG system, companies are financially rewarded with lower financing constraints for their excellence in a wide range of aspects such as environmental protection and social responsibility, providing them with an internal incentive to improve their ESG performance. There are several major difficulties in establishing a market-based ESG system. The first is how to design more mature disclosure standards that balance equity and efficiency to guide companies' scientific ESG disclosure, which is the focus of our research.

The primary and secondary relationship between equity and efficiency has been discussed above, and one of our core recommendations is to improve the efficiency of GRI rather than the equity of SASB. So, as far as the internal designers of information disclosure standards are concerned, and in particular the GRI standard designers, it needs to first accelerate the design of industry-specific standards so that companies can refer to industry-specific standards for ESG disclosure and management as soon as possible. Secondly, in the process of revising the standards, it can consciously mark the possible impact of the required disclosure items on the financial and economic situation of the company, so that companies can make more focused ESG disclosures and capital providers can more easily access the information they need.

The second challenge in establishing a market-based ESG system is the effective identification of ESG information by capital providers, not only in terms of their understanding and receipt of the information, but also in terms of its impact on their investment decisions. While our empirical study on Hypothesis 2 has initially verified the efficiency of direct investors in identifying ESG information, our empirical study on Hypothesis 3 shows that indirect investors, represented by banks, cannot identify ESG information well. Therefore, another core suggestion is for banks to take into account the overall ESG performance of the company rather than just the financial position when lending.

5.4 Deficiencies and Prospects

Due to time constraints, our research design also has certain deficiencies. Firstly, we have not been able to construct a mathematical model at a theoretical level to derive the inevitability of eventual alignment across disclosure standards, nor have we been able to define the two key features of ESG disclosure standards, equity and efficiency, in mathematically rigorous terms, which are the major focus of our subsequent work. Secondly, the empirical analysis of the

fourth hypothesis in this paper, namely “improvement of ESG itself”, is not comprehensive enough, with only one indicator, carbon emissions, being considered. It would also be a direction for subsequent work to determine the differences between GRI and SASB in this respect through more environmental and social indicators. Finally, we have consulted many scholars and third-party institutions to assist us to formulate insights. Yet we did not have an in-depth exchange with professionals who design standards such as GRI and SASB to verify or deepen our understanding of the core differences between them and their future trends. We will contact them as soon as possible when a certain opportunity is available.

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Appendix

Comparison of main ESG Disclosure Standards (Detailed version)

Table 14: Comparison of main ESG Disclosure Standards (Detailed version)

Panel A:Basic Information								
Standards	Standard-setting Organization	Reference Ratio	Year of the First Version	Year of the Latest Version	Count of Revision	Target Audience	Scope of materiality	
GRI	Global Reporting Initiative (GRI)	95%	2000	2020	4	Multiple Stakeholders.	Financial, Environmental, Social	
SASB	Sustainability Accounting Standards Board (SASB)	78%	2018	2018	0	Capital Providers	Financial	
IIRC	International Integrated Reporting Council (IIRC)	75%	2013	2021	1	Capital Providers	Financial	
CDP	CDP Global Environmental Information Research Center	67%	2003	N.A.	N.A.	Multiple Stakeholders	Financial, Environmental, Social	
TCFD	Financial Stability Board (FSB)	57%	2017	2017	0	Capital Providers	Financial	
CDSB	Climate Disclosure Standards Board (CDSB)	30%	2010	2013	1	Capital Providers	Financial	
ISSB	International Accounting Standards Board (ISSB)	N.A.	N.A.	N.A.	N.A.	Capital Providers	Financial	

Note: This table shows the basic information of seven main ESG disclosure standards from seven aspects. The reference ratio is the corresponding reference percentage based on stock exchange guidance on ESG reporting at 2021/12/31, according to the data from the SSE initiative, which is a UN Partnership Programme organized by UNCTAD, the UN Global Compact, UNEP FI and the PRI.

Panel B: Range, Applicability and Goal			
Standards	Scope of Disclosure Topics	Cross-industry Applicability	Main Purpose
GRI	Economy, Environment, Social, Governance	General and Topic standards are cross-industries applicable, while industry standards under developed	Set standards for economic/environmental/social related indicators and disclose them as sustainability reports.
SASB	Environment, Social, Governance	Industry Featured	Disclose financially significant sustainable information for investors
IIRC	Economy, Environment, Social, Governance	Cross-industry applicable	Reveal financial and non-financial correlations to illustrate long-term competitiveness
CDP	Environment	Cross-industry applicable, complemented industry-specific contents	Make companies and cities aware of the environmental impact in order to achieve a sustainable economy
TCFD	Climate-specific	Cross-industry applicable, complemented industry-specific contents	Disclose information on risks/opportunities related to climate change to financial market participants
CDSB	Environment	Cross-industry applicable	Integrate environmental information into financial information and support investment decisions
ISSB	General disclosure (environmental, social, governance) and climate specific	General disclosures is cross-industries applicable, while climate related disclosures is industry featured	N.A.

Note: This table shows the range, applicability and purpose of seven main ESG disclosure standards.

Panel C: Structure, Topics and features			
Standards	Core Issues	Structural Framework	Highlight Features

GRI	3 specific issue categories: Economic, Environmental and Social	General, Topics (2016 version) General, Industries, Topics (2020 version)	The framework of the report is more modularized and clear, consists of economic, environmental and social aspects.
SASB	Environment, Social Capital, Human Capital, Business Models and Innovation, Leadership and Governance	6 elements of General disclosure guidance, Industry descriptions of sustainability topics and Descriptions, Sustainability accounting standards, Technical protocols and Activity metrics	Materiality Map covering 77 industries helps improving the comparability of corporate sustainability data.
IIRC	7 guiding principles and 9 content elements	External Environment, Governance, Business Model, Risks and Opportunities, Strategy and Resourcing, Performance, Outlook, Basis of Preparation, Common Reporting Guidelines	Principle-oriented. Striking the right balance between flexibility and prescription. Recognizing that there are significant differences between the individual circumstances of different institutions, while allowing sufficient comparability across institutions.
CDP	Climate Change Questionnaire, Forest Questionnaire, Water Security Questionnaire	Consists of 3 modules, independently structured standard documents	Help convincing companies to measure, manage, disclose and ultimately reduce their greenhouse gas emissions.

TCFD	4 core elements of Governance, Strategy, Risk management, Metrics and Targets	4 major sections: Recommendations, Recommended disclosures, Industry general guidelines, and Industry-specific guidelines	Comprehensive and systematic. Applicable to all types of organizations in all countries. Promoting the principles of commonality and differentiation of practices related to social responsibility, and is not applicable for third-party certification.
CDSB	Climate Risks and Opportunities	Natural capital reliance, Environmental outcomes, Environmental risks and opportunities, Environmental policies, Strategies and objectives, and Performance against environmental objectives	Adapt rather than expand the organization's main reports and intersperse environmental information in relevant sections of the main reports to explain the linkages between the organization's strategy and environmental performance.
ISSB	N.A.	N.A.	N.A.

Note: This table shows the characteristics of seven main ESG disclosure standards' structure and topics.

Detailed Scoring of GRI and SASB Standards from Section Perspective

Table 16: Detailed Scoring of GRI and SASB Standards from Section Perspective

Serial Number	Section	Sub-section	GRI 2020	SASB 2018
1.1	GOV	Company Purpose (Vision, Mission)	3	1
1.2	GOV	Operationalizing ESG/S	3	1
1.3	GOV	Commitments (by Board and/or C-Suite)	2	1
1.4	GOV	Long-Term Viability of Core Business(es)	3	1
1.5	GOV	ESG/S in Key Business Decisions	3	1
2.1	GOV	CEO/C-Suite Public Posture re ESG/S	3	1
2.2	GOV	CEO/C-Suite Messages to Investors	3	1
2.3	GOV	CEO/C-Suite Interaction: Customers	3	1
2.4	GOV	CEO/C-Suite Messages to Employees	3	1
2.5	GOV	CEO/C-Suite Engagement: NGOs	3	1
3.1	GOV	Full Board Oversight of ESG/S	3	1
3.2	GOV	Board Committees, Charters, and Roles	3	1
3.3	GOV	Board Independence and ESG/S Expertise	3	1
3.4	GOV	Board Agendas Regarding ESG/S	3	1
3.5	GOV	Time Spent on ESG/S in Board Meetings	2	1
3.6	GOV	Board Diversity, Equity & Inclusion	3	1
4.1	GOV	ESG/S Goals and Roadmap: Near-Term	3	3
4.2	GOV	ESG/S Goals and Roadmap: Long-Term	2	1
4.3	GOV	Materiality Assessment	3	1
4.4	GOV	Tracking Reduction of ESG/S Footprint	3	3
4.5	GOV	Revenue: ESG-Advantaged PSS	2	2
4.6	GOV	Accounting for Material ESG/S Risks	1	2
4.7	GOV	ESG/S Ratings and Rankings	2	1
5.1	GOV	Driving ESG/S Culture: Compensation & Goals	3	1
5.2	GOV	ESG/S Organization	3	1
5.3	GOV	ESG/S Leadership Role/Position	1	1
5.4	GOV	ESG/S Reward and Recognition	1	1
6.1	GOV	Annual Reporting & Financial Disclosures	3	3
6.2	GOV	Disclosure of ESG/S Impacts and Strategy	3	3
6.3	GOV	Assurance and Verification of ESG/S	3	1
6.4	GOV	Transparency	3	1
6.5	GOV	Public Policy (e.g., Lobbying) Alignment	2	1
6.6	GOV	Corporate Marketing and Advertising	2	1

Serial Number	Section	Sub-section	GRI 2020	SASB 2018
7.1	STR	Customer ESG/S Engagement	2	1
7.2	STR	Strategy: Product ESG/S Features	1	1
7.3	STR	Market Expansion: ESG Features	1	1
7.4	STR	ESG/S in Product Line Extension	1	3
7.5	STR	ESG/S Impacts on Brand	1	1
8.1	STR	Strategic Planning Process	3	2
8.2	STR	Use of Scenario Planning	1	2
8.3	STR	ESG/S Cost Reduction	1	3
8.4	STR	Enterprise Risk Management	3	3
8.5	STR	Revenue Pipeline: ESG/S Products	1	3
8.6	STR	Capital Allocation: ESG/S Impacts	1	3
9.1	STR	Linkage: ESG/S & Innovation	2	3
9.2	STR	ESG/S in Materials and Labor Inputs	3	3
9.3	STR	Product Design & Development	1	3
9.4	STR	R&D Partnerships: ESG/S Benefits	1	1
9.5	STR	ESG/S Investment: PSS & Technology	1	2
10.1	STR	Product (PSS) Value Proposition	1	3
10.2	STR	Product (PSS) Stewardship	2	3
10.3	STR	Product ESG/S Risk Assessment	1	2
10.4	STR	Product ESG/S Labeling & Rating	2	3
10.5	STR	Product (PSS) Quality and Safety	2	3
10.6	STR	PSS Marketing & Advertising	2	1
11.1	STR	Responsible Sourcing Approach	3	3
11.2	STR	Engaging on Material (SC) Issues	3	2
11.3	STR	Standards for SC ESG/S Impacts	2	2
11.4	STR	Measuring SC ESG/S Impacts	3	2
11.5	STR	Verifying SC ESG/S Impacts	3	2

Serial Number	Section	Sub-section	GRI 2020	SASB 2018
12.1	ENV	GHG Emissions: Scope 1	3	3
12.2	ENV	GHG Emissions: Scope 2	3	3
12.3	ENV	Non-Carbon Emissions	3	3
12.4	ENV	Buildings and Equipment	2	3
12.5	ENV	Water Management	3	3
12.6	ENV	Biodiversity and Land Management	3	2
12.7	ENV	Waste Management	3	2
13.1	ENV	Approach: Supply Chain Env'l. Impacts	3	2
13.2	ENV	Materials Sourced: Human Made	3	3
13.3	ENV	Materials Sourced: Biological-based	2	2
13.4	ENV	GHG Emissions: Scope 3	3	1
13.5	ENV	Supply Chain Impact: Biodiversity	3	1
13.6	ENV	Supply Chain Impact: Water	3	1
13.7	ENV	Supply Chain Impact: Waste	2	2
14.1	ENV	Responsibility: Product Use, End-of-life	3	3
14.2	ENV	Traceability Related to Products (PSS)	1	2
14.3	ENV	Carbon (GHG) Impact of Products (PSS)	2	3
14.4	ENV	Durability of Products (PSS)	2	2
14.5	ENV	Biodegradability of Products (PSS)	1	1
14.6	ENV	Recyclability and Reusability of Products	3	3
14.7	ENV	Water-Use Efficiency of Products (PSS)	3	3
14.8	ENV	Product Packaging	2	1

Serial Number	Section	Sub-section	GRI 2020	SASB 2018
15.1	SOC	Workplace Environment	2	3
15.2	SOC	Diversity, Equity, and Inclusion (DEI)	3	1
15.3	SOC	Recruitment and Retention	1	1
15.4	SOC	Safety Programs and Performance	3	3
15.5	SOC	Health, Wellness, and Lifestyles	1	2
15.6	SOC	Training and Staff Development	1	2
15.7	SOC	Employee Engagement	1	1
16.1	SOC	Approach: Supply Chain Social Impacts	3	2
16.2	SOC	Human Rights	3	3
16.3	SOC	Labor Relations	3	2
16.4	SOC	Animal Welfare	1	1
16.5	SOC	Supply Chain Capacity Building	2	2
17.1	SOC	Philosophy re “Community”	3	3
17.2	SOC	Engagement with Communities/NGOs	3	2
17.3	SOC	Social Investment	2	2
17.4	SOC	Stakeholder Partnerships	1	3
17.5	SOC	Infrastructure Development	3	3
17.6	SOC	Community Job Creation	1	3

Note: This table shows the corresponding difference of specific disclosure of three comparable companies’ ESG Reports in 2021 with detailed company profile stated above, in accordance with the characterization of the requirements in Panel A.

Word Bank by Sections

- ENVIRONMENT:

environment, climate, air, water, biodiversity, ecosystem, sustainable, tree, emission, GHG, greenhouse, carbon, biological, biodegradability, chemical, nature, natural, oxygen, hydro, ozone, pollution, renewable, sewage, solar, recycle, eco-friendly, pollute, recycling, forest, landfill, global warming, sustainability, organic, pollutant, exhaust fumes, humidity, soil, atmosphere, atmospheric, durability

- SOCIAL:

society, social, safety, workplace, security, community, communities, worker, employee, workforce, infrastructure, stakeholder, NGO, human rights, staff, labor, recruitment, diversity, inclusion, animal welfare, social investment, job creation, create jobs, volunteer, non-profit, responsibility, responsibilities

- GOVERNANCE:

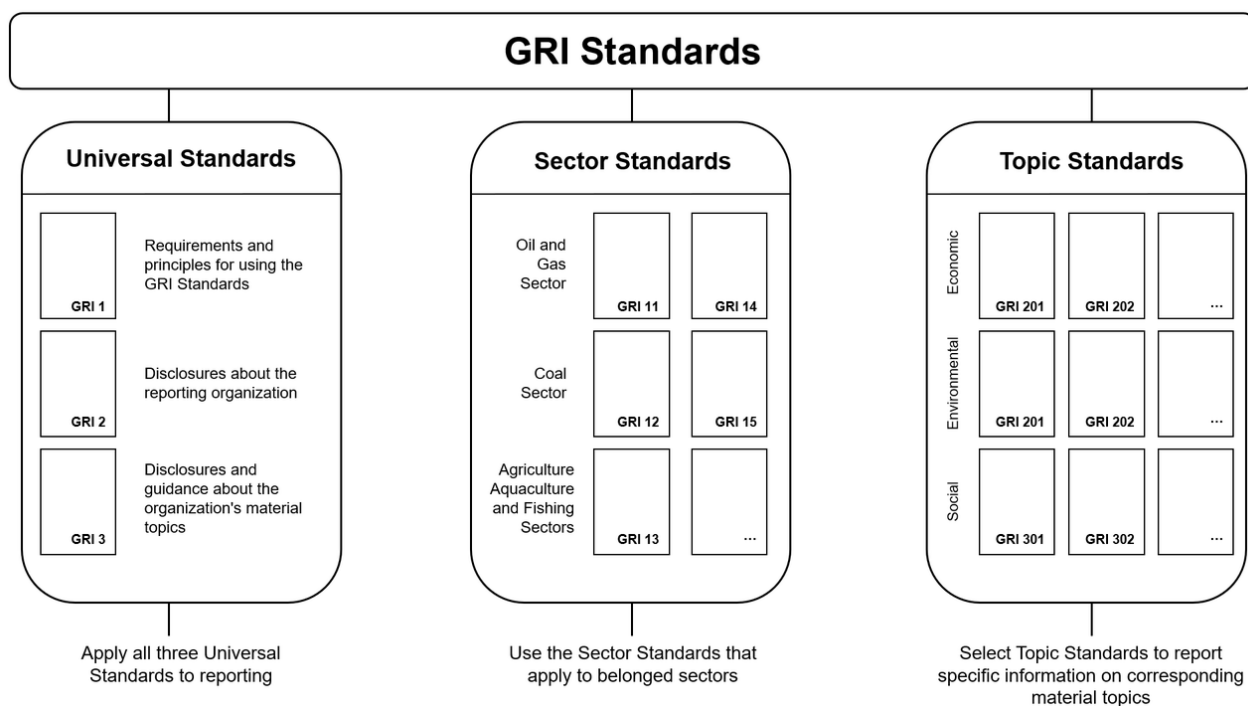
ethics, integrity, board, shareholder, govern, C-suite, CEO, committee, compensation, reward, leadership, company culture, corporate culture, stewardship, materiality, independence, roadmap

- STRATEGY:

strategy, economic, equity, asset, debt, liability, liabilities, revenue, profit, sales, cost, risk management, capital, R&D, strategic, technology, technologies, advertise, marketing, advertising, risk assessment, finance, financial, equities, value proposition, research, development, innovate, innovation, product design, market expansion, market share, product line, industrial chain, supply chain

Comparison between the structure of GRI and SASB Standards

Figure 8: The Structure of GRI Standards



Note: The figure shows the structure of the latest GRI2020 Standards, which provides guidance of disclosure methodology for any company adopting GRI Standards.

Figure 9: The Structure of SASB Standards (SICS®)

SASB Standards	Thematic Sectors	Sub-Sectors	Industries
	Consumer Goods	Apparel & Textiles ...	Apparel, Accessories & Footwear / Appliance Manufacturing / Building Products & Furnishings / Household & Personal Products / Toys & Sporting Goods / Multiline and Specialty Retailers & Distributors / E-commerce
	Extractives & Minerals Processing	Construction Materials ...	Coal Operations / Construction Materials / Iron & Steel Producers / Metals & Mining / Oil & Gas - Exploration & Production / Oil & Gas - Midstream / Oil & Gas - Refining & Marketing / Oil & Gas - Services
	Financials	Capital Markets ...	Asset Management & Custody Activities / Investment Banking & Brokerage / Security & Commodity Exchanges / Commercial Banks / Consumer Finance / Mortgage Finance / Insurance
	Food & Beverage	Food ...	Agricultural Products / Meat, Poultry & Dairy / Processed Foods / Alcoholic Beverages / Non-Alcoholic Beverages / Food Retailers & Distributors / Restaurants / Tobacco
	Health Care	Medical Technology ...	Biotechnology & Pharmaceuticals / Drug Retailers / Health Care Delivery / Health Care Distributors / Managed Care / Medical Equipment & Supplies
	Infrastructure	Utilities ...	Electric Utilities & Power Generators / Gas Utilities & Distributors / Water Utilities & Services / Engineering & Construction Services / Home Builders / Real Estate / Real Estate Services / Waste Management
	Renewable Resources & Alternative Energy	Alternative Energy ...	Biofuels / Fuel Cells & Industrial Batteries / Solar Technology & Project Developers / Wind Technology & Project Developers / Forestry Management / Pulp & Paper Products
	Resource Transformation	Chemicals ...	Aerospace & Defense / Containers & Packaging / Electrical & Electronic Equipment / Industrial Machinery & Goods / Chemicals
	Services	Media ...	Advertising & Marketing / Media & Entertainment / Casinos & Gaming / Hotels & Lodging / Leisure Facilities / Education / Professional & Commercial Services
	Technology & Communications	Technology ...	Electronic Manufacturing Services & Original Design Manufacturing / Hardware / Software & IT Services / Internet Media & Services / Semiconductors / Telecommunication Services
	Transportation	Air Transportation ...	Airlines / Air Freight & Logistics / Automobiles / Auto Parts / Car Rental & Leasing / Cruise Lines / Marine Transportation / Rail Transportation / Road Transportation

Note: The figure shows the structure of the SASB2020 Standards, which provides guidance of disclosure methodology for companies differing in industries and adopting SASB Standards.

Word Bank by Mandatory Level

-REQUIREMENTS:

shall, must, need, require, requiring, requirement, requirements, required, ought, have to, have got to, obliged, compel, obligation, necessity, necessary, essential

-RECOMMENDATIONS:

should, encourage, encourages, encouraging, encouraged, encouragement, recommend, recommended, recommendation

-OPTIONS:

can, could, may, might

Case Analysis based on Original Text of the GRI and SASB Standards

Table 17: Case Analysis based on Original Text of the GRI and SASB Standards

PanelA: Comparison between Specific Indicators of GRI and SASB Standards				
Characteristics	Topic	ESG Disclosure Standards		
		GRI (2016)	GRI (2020)	SASB (2018)
Scope of Indicator Alternatives	GHG Emission	GRI305-4: The reporting organization shall report the gases included in the calculation; whether CO ₂ , CH ₄ , N ₂ O, HFCs, PFCs, SF ₆ , NF ₃ , or all	GRI305-4: Gases included in the calculation; whether CO ₂ , CH ₄ , N ₂ O, HFCs, PFCs, SF ₆ , NF ₃ , or all	SASB-Oil exploration: 7 GHGs covered under the Kyoto Protocol, which are CO ₂ , CH ₄ , N ₂ O, HFCs, PFCs, SF ₆ , and NF ₃
	Air Emission	GRI305-7: NO _x , SO _x , POP, VOC, HAP, PM, other standard categories of air emissions identified in relevant regulations	GRI305-7: NO _x , SO _x , POP, VOC, HAP, PM, other standard categories of air emissions identified in relevant regulations	EM-EP-120a.1. Air emissions of the following pollutants: (1) NO _x (excluding N ₂ O), (2) SO _x , (3) VOCs, and (4) PM ₁₀
Classification of Indicators	Water	GRI303-3: Can be classified by Surface water, Ground water, Seawater, Produced water, Third-party water	GRI303-3: Can be classified by Surface water, Ground water, Seawater, Produced water, Other water	EM-EP-140a.1: Provide definition of Fresh water and only divide water into Fresh water, Ground water and Surface water
	Waste	GRI306-3: Material of spill, categorized into: Oil spills (soil or water surfaces), Fuel spills (soil or water surfaces), Spills of wastes (soil or water surfaces), Spills of chemicals (mostly soil or water surfaces), and Other (to be specified by the organization).	N.A.	EM-EP-160a.2 May disclose spills to soil and water separately
Definition of Indicators	Waste	N.A.	N.A.	EM-EP-160a.2: Spills are defined as greater than 1bbl (42 U.S. gallons or 159 liters). Spills include those that reached the environment and exclude spills that were contained within impermeable secondary containment
Perspective of Disclosure	Work and Safety	GRI403-Occupational Health and Safety/404-Training and Education/408-Child Labor/409-Forced or Compulsory Labor	GRI403-Occupational Health and Safety/404-Training and Education/408-Child Labor/409-Forced or Compulsory Labor	EM-EP-210:Security, Human Rights & Rights of Indigenous Peoples/EM-EP-320: Workforce Health & Safety

Note: This table shows the difference of specific required indicators of GRI and SASB standards, covering topics of GHG emission, air emission, water, waste work and safety.

PanelB: Comparison between Corresponding ESG Reports from Comparable Companies				
Information about Comparable Companies				
Name		APA	OXY	COP
ESG Disclosure Standards Used		GRI 2016	SASB 2018	GRI 2016 & SASB 2018
Company Profile		APA Corp. explores for oil and natural gas through its subsidiaries. It produces oil and gas with operations in the United States, Egypt and the United Kingdom, and exploration activities offshore in Suriname. The company was founded in 1954 and is headquartered in Houston, TX.	Occidental Petroleum is the fourth largest oil and gas company in the United States, engaged in the exploration and production of oil and natural gas and trading assets around transportation and storage capacity. The company was founded in 1920 and is headquartered in Houston, Texas.	ConocoPhillips International, Inc. is the world's largest independent exploration and production (E&P) company, founded in 1875 and headquartered in Houston, Texas, USA.
Industry		Oil, Gas & Consumable Fuels	Oil, Gas & Consumable Fuels	Oil, Gas & Consumable Fuels
Major Businesses		The Exploration and Production of crude oil and natural gas	Oil and Gas, Chemicals, Midstream and Marketing.	Exploration, Production, Transportation and Sale of crude oil, bitumen, natural gas, LNG and natural gas liquids worldwide
Market Capitalization		\$13.75 Bn	\$60.32 Bn	\$148.74 Bn
Specific Disclosure Indicators in each 2021 ESG Report				
	Name	APA	OXY	COP
Scope of Indicator Alternatives	GHG Emission	Carbon dioxide (CO2), Methane (CH4) and Nitrous oxide (N2O)	7 GHGs including CO2, CH4, N2O, SF6, PFCs, HFCs, NF3	CO2, CH4, N2O, and 3 refrigerant gases (CFCs, HCFCs, and HFCs)
	Air Emission	PM2.5, PM10, SO2, NOx, VOCs, CO, Pb	NOx, SOx, CO, VOCs, PM, HAPs, ODS	N.A.
Classification of Indicators	Water	Divide water into Recycled produced water, Non-fresh water and Fresh water	Divide water into Fresh water and Non-fresh water	Divide water into Surface water (including water from Wetlands, Rivers, Lakes and Ocean), Ground water, Rainwater, Waste water
	Waste	Produced liquids spilled, solid spills, oil spills	Reportable spills, Produced water or Crude	N.A.
Definition of Indicators	Waste	N.A.	Reportable events include spills or releases in excess of a level stipulated by regulatory agencies or a deviation from a regulatory standard or an established permit condition	N.A.

Note: This table shows the corresponding difference of specific disclosure of three comparable companies' ESG Reports in 2021 with detailed company profile stated above, in accordance with the characterization of the requirements in Panel A.